



UMEÅ UNIVERSITY

Monthly Electricity Consumption Forecasting Using an Explainable AI Framework

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Abstract

Checklist for final touches:

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1 Introduction

In recent years, the Swedish electricity market has been affected by large variations in price and periodically high electricity costs¹². This has increased the need for customers to understand and influence their electricity consumption and future electricity costs.

With the emergence of environmental problems due to climate change, sustainable energy management is becoming increasingly important [1, 2]. The ability for customers to understand and influence their household energy consumption is significant for sustainable energy use. Household energy consumption makes up a considerable part of the total energy consumption, approximately 30% in some European and American countries. Managing household energy more efficiently has large potential in saving energy, since it is estimated that 27% of energy used by households can be saved through more efficient use [3].

[Something about how the energy demand in the world is increasing]

[Something about how forecasting electricity consumption can help users make smart decisions about their consumption, both saving money and benefitting the environment.]

[Something about monthly forecasting and why that is useful]

For users, the ability to have confidence in the forecast is important. Users may, for example, question how the forecasting mechanism makes decisions, or question what factor is the most determining in the forecast. A system that can explain such things to the user is able to increase its reliability with users. Handling users' data in a safe and transparent manner is also important when implementing Artificial Intelligence (AI) based systems. Citizens of the European Union have a right to transparency and information about the decision-making of AI models that have a direct link to them, as outlined in the General Data Protection Regulation [4]. [Something about how XAI helps with these things]

Previous research has identified various models for predicting electricity consumption, both statistical and AI based [5, 6]. In recent years, increasing interest has been shown for AI solutions to energy consumption forecasting, due to their strength in handling nonlinear problems [4, 5]. Forecasts are often based on several factors such as weather conditions and building conditions, causing nonlinear patterns that statistical methods struggle with [5]. However, one disadvantage with AI methods is the black-box nature of the models which causes a lack of transparency and interpretability in the decision-making process. To solve the problem of elucidating the internal process and decision-making of AI algorithms, the field of explainable AI (XAI) has emerged within recent years. XAI methods are able to open up the black box, and provide explanations for why a model predicts a certain way [7].

¹Allhorn, J., Tidningen Näringslivet: Chockvändningen: Nu har norra Sverige högst elpriser - "Har förändrats ganska snabbt" (January 2026). <https://www.tn.se/inrikes/46369/chockvandningen-nu-har-norra-sverige-hogst-elpriser-har-forandrats-ganska-snabbt/>

²Rydegran, E., Energiföretagen: Dramatik och rekord sammanfattar Elåret 2022 (December 2022). <https://www.energiforetagen.se/pressrum/pressmeddelanden/2022/Dramatik-och-rekord-sammanfattar-Elaret-2022/>

[To address this need, Umeå Energi has worked on developing and improving their interfaces for customers, for example by developing an application for private customers. According to customer analytics from Umeå Energi, a highly requested feature is to receive an estimation of both the electricity consumption and the electricity cost before the invoice arrives. Currently, Umeå Energi has developed a linear regression model to forecast the consumption for the current month. The model is based on historical consumption and temperature, aggregated by month. For certain customer groups, Umeå Energi has access to high definition measurement data where customer electricity consumption is registered on a 15-minute level, which creates opportunities for more advanced modeling. There may also be other factors, apart from historical consumption, that affect the electricity consumption, for example weather data, calendar information (holidays etc.), and energy prices.]

[This Master's thesis will investigate and quantify the contributions of different variables to a monthly electricity consumption forecast. Previous research has identified several models that are able to accurately predict energy consumption for different time intervals, utilizing various combinations of data sources [6, 5]. In recent years, increasing interest has been shown for Machine Learning (ML) and AI solutions to energy consumption forecasting, due to their strength in handling nonlinear problems [4, 5]. However, the black-box property of many popular AI models has increased the demand of asserting confidence and transparency through means of Explainable Artificial Intelligence (XAI) [4]. This study will address this demand by using XAI methods to evaluate how different variables such as weather data, holidays, and energy prices affect the accuracy of monthly household electricity consumption forecasts. Previous research has used XAI approaches for short-term forecasting (hourly-daily), but this study will explore a novel monthly scope [2, 8, 4, 9]. The access to customer-level data in this study also provides an aspect that is underrepresented in previous works [5, 10, 9]. **RQ1: Which forecasting variables contribute significantly to the accuracy of the monthly forecast, and which variables provide little to no improvement?**

To perform the forecasting, state-of-the-art ensemble learning models will be explored. Previous studies have used ensemble learning for short-term (hourly-daily) forecasting [11, 12, 9], but this study will address the novel task of monthly forecasts using ensemble learning. **RQ2: Which ensemble learning models are the most efficient for forecasting monthly electricity consumption?**]

For Umeå Energi, this research will add value by supporting customers in making informed decision about their energy consumption. According to internal customer analytics, a highly requested feature is to receive an estimation of both the electricity consumption and the electricity cost before the invoice arrives. Umeå Energi is currently in the process of launching an application for private customers. The ultimate goal is to include to monthly consumption forecasts in the application.

[Definition of short/mid/long-term forecasts]

[The contributions of this work can be summarized as follows:]

2 Background

This chapter presents theoretical background for concepts and methodologies used in this thesis, including an introduction to time series forecasting, ensemble learning models, and explainable artificial intelligence. The chapter also presents previous works related to this thesis, with a focus on works utilizing ensemble learning techniques and explainable artificial intelligence for energy consumption forecasting.

3 Theory

This chapter presents relevant theoretical background for the concepts and methodologies used in this research.

3.1 Time Series Forecasting

A time series is a set of chronologically arranged observations, generally under the assumption that time is a discrete variable [13]. In a time series with T real value samples $x_1, \dots, x_T$, the value at time i is represented by x_i ($1 \leq i \leq T$). The problem of time series forecasting can be defined as predicting the values of $x_{w+1}, \dots, x_{w+h}$, where w is the historical window, and h is the prediction horizon. The historical window corresponds to the number of past observations used to generate the forecast, and the prediction horizon defines the number of future time steps to be predicted. Prediction happens given the previous samples $x_1, \dots, x_w$ ($w + h \leq T$), with the goal of minimizing the error between the predicted value $\hat{x}_{w+i}$ and the actual value x_{w+i} ($1 \leq i \leq h$) [11].

A time series may be univariate or multivariate, where univariate refers to a series with a single observation recorded over time, and multivariate refers to a series with a group of variables that may exhibit interactions. Analysis of time series often begins with obtaining the underlying patterns of the observed data. The next step is fitting a model to represent the data for future predictions, which can be a complex task. The analysis of the time series is usually performed by decomposing it into three components [6, 14]:

1. *Trend* - The general movement of the variable during the observation period. Trend does not take irregularities and seasonality into account.
2. *Seasonality* - The periodic fluctuation of the variable. It includes stable effects, along with time, magnitude, and direction.
3. *Residual* - The remaining variation, a largely unexplainable part of the time series. In some instances, this can be high enough to mask the trend and seasonality.

Both linear and nonlinear techniques may be used for solving time series problems. In linear methods, the idea is that strong correlations in the data allow for linear combinations to determine the next observation. However, the random component of the time series may prevent precise predictions. Nonlinear methods are applied in the machine learning domain to forecast future values based on a model describing the data. Machine learning solutions to forecasting problems have gained popularity, due to the fact that they are suitable both for linear and nonlinear problems [11]. Time series analysis has been extensively applied to the problem of energy consumption forecasting, especially since real-time monitoring of buildings has become more common, and the availability of recorded data has increased [6].

Based on the forecasting horizon, energy consumption forecasting problems are generally divided into three categories; short-term forecasting (1 hour to 1 week), medium-term forecasting (1 month to 1 year), and long-term forecasting (1 year and above) [5]. This thesis explores medium-term forecasting with a horizon of 1 month.

3.2 Ensemble Learning

Ensemble learning is a family of methods that combine several machine learning algorithms to make decisions. It can be seen as the ML interpretation of “wisdom of the crowd”, that is, the idea that aggregating the opinions of several individuals is better than selecting the opinion of one individual. Multiple weak learners make predictive results that are fused together via voting mechanisms, in order to achieve better performance than from any single algorithm. Any type of machine learning algorithm, e.g. decision trees, neural networks, or regression models, can be used as ensemble learners [15, 16].

[maybe a diagram giving an overview of the ensemble “steps”]

According to Dietterich [17], there are three reasons why ensemble learning methods achieve good performance in machine learning. The first reason is statistical. Models can be seen as searching for the best hypothesis within a hypothesis space H . When data is limited, which it often is, the models can find several hypotheses in H that give the same accuracy on the training data. This makes it hard to choose between hypotheses, since it is unknown which one will generalize better for unseen data. Ensemble learning can help with avoiding this problem of overfitting. Secondly, ensemble learning provides computational benefits. Many models function via performing some type of local search, which may be prone to getting stuck in a local optima. A better approximation of the unknown true function may be obtained from an ensemble constructed by the local search started from many different points. The third reason is representational. In most situations, the unknown true function may not be included in H . Combining several hypotheses from H , the space of representable functions can be expanded to possibly include the true function.

Three common strategies for constructing ensembles are bagging, boosting, and stacking [9]. Bagging, or bootstrap aggregating, proposed by Breiman [18] is a method for obtaining an aggregated prediction by building multiple versions of a predictor. Bootstrapping is used to generate multiple samples of data from the original dataset, which are then used to independently train the different predictors. Majority voting is used to aggregate the final prediction [15]. The main objective of the bagging technique is to reduce the variance of a prediction model. A common example of the bagging technique is the random forest (RF) algorithm [9, 19].

In boosting, several models with individually weak predictions are aggregated to obtain a model with high accuracy. Models are trained sequentially to correct the errors of previous models. The main objective with this technique is reducing model bias [9, 19, 20]. Two examples of boosting algorithms are Gradient Boosting Machines (GBM), which uses gradient descent to minimize the loss function, and XGBoost, which is an extension of GBM focused on resource optimization and prevention of overfitting [19].

Stacking utilizes different learning algorithms called base learners to make initial predictions. These outputs are then combined via a meta-learner to make the ultimate predictions.

The goal is to capture a wider range of patterns using the strength of different models, while minimizing the generalization errors [9, 19, 11]. Accuracy improvements mainly happen when there is diversity among the individual models, since differences in generalization principles generally lead to different results [19].

3.3 Explainable Artificial Intelligence

In recent years, artificial intelligence has been widely adopted in society, and humans increasingly rely on AI as part of decision-making. Research in the field of AI has for many years focused on improving the predictive accuracy. However, the black-box nature of many AI systems creates a lack of transparency, and explaining to users how the AI systems work is highly problematic. Pressure from society, ethics, and legislation has created a demand of a new type of AI that is interpretable to users and can explain its inner functions [7, 21].

Machine learning models have two main objectives; firstly, they should be able to make accurate predictions for new given input features, and secondly, they should be interpretable (i.e., provide explanations for how a given input relates to the output). Usually, models have a trade-off between accuracy and interpretability. Linear regression models are generally transparent and easy to understand, but their performance is often worse than more complex nonlinear models. On the other hand, the nonlinear models perform well but lack in interpretability [22].

Explainable AI (XAI) methods that elucidate the decision-making process have been developed to enhance the interpretability of AI models. These methods can be applied at different levels of the AI modelling. Firstly, explainability can be applied pre-modelling through means of data pre-processing and data analysis. This approach gives insights into the datasets used for training the ML models, and it can expose imbalances in the data. Secondly, the model itself can be inherently interpretable. This refers to models that have a simple structure, such as linear regression and k-Nearest Neighbors. Models like these are understandable by design, and users are able to interpret the model based on the model summary or parameters. Thirdly, explainability can be applied post-modelling to deal with the black-box problem of more complex ML models. Post-modelling methods include textual and visual justifications, simplification of the model, and feature relevance. These methods can explain how an algorithm performs during training, and how predictions are generated given a certain input [7, 21].

XAI approaches may be *model-specific*, meaning they are applied to a specific type of model, or *model-agnostic*, meaning they are not dependent on model architecture and can be applied for any model. Additionally, XAI explanations can be *local*, which means that individual predictions from given feature values are explained, while *global* explanations mean that feature contributions are explained for all instances [21, 23].

Feature importance is the most common explanation tool for classification models [22]. It describes how important a variable was for the model output, that is, the individual contribution of a feature, regardless of the shape or direction of the feature's effect, dependent on a particular classifier. Feature importance reveals which features are crucial for the model when making predictions. The feature importance descriptions may be local or global, where local importance describes the feature's contribution to a specific prediction, and global importance describes the feature's contribution for the entire model [22, 21, 7].

Ensemble models, used in this thesis, are based on the aggregation of several models, which makes it more complex to interpret the results. Since the models are not interpretable by design, there is a need for the implementation of post-modeling XAI methods to understand the models' predictions. Among the post-modeling XAI methods, simplification and feature relevance are the most widely used for this task [7].

3.3.1 SHAP

SHapley Additive exPlanations (SHAP) is a popular method for model explainability. It is model-agnostic and can therefore be used for any ML model. The SHAP method has its foundation in game theory, where the contribution of each player to the total payoff is calculated. In the case of machine learning, the players are represented by features (variables) and the payoff is the model output. To calculate the contribution score for each feature, all different combinations of features (i.e., coalitions in game theory) are evaluated. The Shapley value of feature i can be calculated according to Equation 3.1, where ϕ_i is the relative contribution to the model output, M is the set of all features, and $f_S(x)$ is the model output for feature subset S [23, 24].

$$\phi_i = \sum_{S \subseteq M \setminus \{i\}} \frac{|S|!(|M| - |S| - 1)!}{|M|!} [f_{S \cup \{i\}}(x) - f_S(x)] \quad (3.1)$$

A positive Shapley value for a feature means that the feature contributes to increasing the predicted value. For example, when predicting electricity consumption, a substantially positive Shapley value for temperature means that the increasing temperature contributes to an increased electricity usage. SHAP can be used to explain model outputs both locally and globally, i.e., both for specific instances and for all instances [9, 23].

When using SHAP for model explanations, it is important that users are aware of some important considerations of the method. Firstly, the SHAP method is model-dependent, meaning it can produce different explanations for the same task when different models are applied. Secondly, the explainability scores assigned to features do not represent the weights of the features. Rather, the explainability lies in the ranking of the features, meaning that feature importance is deduced from the ordering. Thirdly, it is assumed that the features are independent of each other. Therefore, some features that are correlated with other features may receive low scores even though they contribute significantly to the output. This is because the correlated features have already accounted for the impact of the low-scoring feature. Lastly, if SHAP is applied to a biased classifier (e.g. due to data bias or algorithmic bias), the method may produce unrealistic explanations that do not describe the bias. Because of these points, it is crucial to present SHAP results in an informative way by including corresponding output plots and assumptions behind the method [23].

4 Previous Work

Research in the field of electricity consumption forecasting has established many methods for producing forecasts. These methods can roughly be divided into two categories; conventional methods and AI-based methods. Conventional methods such as time series analysis and regression methods have previously been widely applied to solve the problem of electricity consumption forecasting. Nowadays, AI-based methods are more popular due to their strength in solving nonlinear problems, something that the conventional methods struggle with [5]. Overall, two of the most widely applied models in the field are artificial neural networks (ANN) and support vector machines (SVM), along with the statistical autoregressive integrated moving average (ARIMA) model [25].

4.1 Monthly Forecasting

Forecasting electricity consumption in the monthly scope requires handling complex nonlinear patterns and seasonal tendencies [26]. Monthly consumption is affected by factors such as economy, temperature, and season, which makes the consumption exhibit a mixture of trend, periodicity, and randomness [27]. Additionally, when predicting the consumption of small-scale users, like in this thesis, the power demand is more sensitive to random factors. Human behavior may change periodically, both due to social factors such as holidays and summer vacations, and due to natural factors like the change in seasons [28].

To address the seasonality of monthly forecasting, models incorporating seasonal indexing (SI) have been proposed. Cao and Wu [26] aimed to improve monthly electricity consumption forecasts using support vector regression (SVR) in combination with SI and a novel evolutionary algorithm called fruit fly optimization (FOA). They calculated the seasonal index for each month, which was then used to adjust the predicted values according to the seasonal tendencies. FOA was applied to find the best values for parameters of the SVR model. Their results showed that both SI and FOA was able to effectively improve the forecasting accuracy, suggesting that the hybrid approach is a viable option for monthly forecasting.

Tang et al. [27] proposed a model specifically designed to handle sparse data with periodic properties. Their model also incorporated SI to address the seasonal component. A GM(1,1) gray model, a mathematical optimization model for incomplete data, was used to predict the seasonal index. The SI was combined with a trend forecast value to obtain the final prediction, and results showed that the model significantly improved accuracy, even for small electricity consumption datasets.

Sun et al. [28] approached the monthly forecasting problem by seasonal and trend decomposition of the electricity load time series, in combination with Auto-regressive Integrated Moving Average (ARIMA). This technique allowed for control of both the seasonal compo-

nent's change rate and the trend component's smoothness, as well as increased robustness to outliers. Additionally, they incorporated economic factors into the forecast to improve accuracy. They found that the proposed model was able to increase forecasting accuracy, and determined that both the decomposition approach and inclusion of economic variables were effective to the forecast.

4.2 Ensemble Learning Approaches

Ensemble learning has previously been applied for the task of electricity consumption forecasting, mostly in the short-term scope. Divina et al. [11] proposed a stacking ensemble learning approach for short-term forecasting of Spain's general electricity consumption. They used a two-layer approach based on a bottom and a top layer. The bottom layer consisted of three base learning methods; regression trees based on evolutionary algorithms, Artificial Neural Network (ANN), and Random Forest (RF). The top layer used Generalized Boosted Regression Models (GBM) to combine the predictions of the base learners, and produce the final predictions. They found that the combination of weaker learning methods was able to produce superior results compared to several baselines, showing that ensemble learning is suitable for short-term electricity consumption forecasting. Neural networks were able to predict with high accuracy in the very near future, but accuracy declined when predicting further in the future. The ensemble method also showed significantly better performance compared to baselines when the amount of historical data was small, indicating that ensemble methods may be more flexible for different quantities of data.

Pinto et al. [12] also utilized ensemble learning for forecasting electricity consumption in the short-term. Using consumption data from an office building, along with other variables such as temperature and humidity, they predicted hour-ahead consumption with three different ensemble methods. These methods were random forests, gradient boosted regression trees (GBR), and AdaBoost. They found that ensemble learning methods, especially AdaBoost, outperformed benchmark models such as support vector regression and fuzzy rule-based methods. They highlight including temperature and humidity information as advantageous to the forecasts, and they suggest that future work should consider additional external variables such as solar irradiation and thermal sensation.

Khan et al. [29] presented a stacking deep learning ensemble for short-term forecasts. The lowest level of the ensemble was comprised of multiple individual Long Short Term Memory (LSTM) models responsible for producing outputs. The top level of the ensemble was a Gated Recurrent Unit (GRU) responsible for increasing accuracy by combining outputs from the previous layer. They also utilized clustering to identify customer groups with similar profiles and load patterns. Based on hourly residential consumption data, they predicted hourly, daily, and weekly consumption, and found that the proposed model outperformed benchmark deep learning models. The clustering of customers helped with reducing model errors. Additionally, it was found that their model performed best for hour-ahead forecasts, while errors increased when approaching the weekly scales.

Hadjout et al. [30] used an ensemble deep learning approach to forecast monthly electricity consumption for high voltage customers in the Algerian economic sector. They utilized three different deep learning models - LSTM, GRU, and Temporal Convolutional Networks (TCN) - and grid search to find optimal weights for each model in the final ensemble pre-

diction. Compared to the individual models separately, the ensemble approach performed significantly better according to MAE and MAPE metrics.

Long-term forecasting using ensemble learning has been researched by Chen et al. [31]. They predicted the annual electricity consumption for American households using extreme gradient boosting forests and deep networks as base learners, and ridge regression as a combiner method. The data included variables pertaining to household demographics, household type, appliances, and geographical data, providing a wide range of information. The ensemble framework was able to reduce errors by 30% compared to classical regression models such as linear regression and support vector machines. Investigation into the importance of input features using principal component analysis showed that electricity price, number of rooms, natural gas usage, and temperature during summer were among the variables that contributed the most information. They also note that some features, like number of rooms, number of bedrooms, and total square footage, have a strong correlation and have potential to be represented more efficiently. They argue that feature selection is necessary, since discarding of unimportant features can enhance model performance.

4.3 Explainable AI Approaches

Explainable AI methods have previously been applied for the task of short-term electricity consumption forecasting. Sim et al. [2] proposed a novel XAI approach for variable selection in energy consumption forecasting. They analyzed the influence of 15 different input variables pertaining to time information, climate data, and past energy consumption data. LSTM was selected as prediction model and SHAP was used to interpret the variables. The findings showed that time data such as year and hour had a strong influence on the prediction results, along with temperature, surface temperature, and energy-consumption-difference. Variables with a weak or ambiguous influence included humidity, wind speed, and sunshine amount. They concluded that forecasting performance can be significantly improved by selecting high-impact variables based on XAI analysis.

Maarif et al. [8] investigated the effect of 9 different features on energy consumption forecasting for a South Korean steel company. The features provided information about historical consumption, lagging and leading current power, carbon dioxide emissions, and load type. They utilized various LSTM models to make hour-ahead forecasts, and they used SHAP for the XAI analysis. It was found that leading current reactive power and seconds from midnight strongly influenced the model output, while CO₂ and load type had no effect on the forecast. They note that it is important to consider interpretability and performance trade-offs when developing XAI forecasting models, and that further research into this topic is needed for understanding the applicability of these models in the energy sector.

Januja et al. [4] maybe, but not reputable journal.

Another hourly electricity consumption forecasting study was performed by Lazzari et al. [10]. They provided a novel perspective by incorporating user behavior to enhance the accuracy of residential energy consumption forecasting. User behavior clusters were identified, and the day-ahead behavior was predicted using an XGBoost model. The predicted behavior, along with historical consumption, was then provided to an ANN model responsible for predicting the hourly day-ahead consumption. Compared to an ANN without the user behavior input, the proposed model was able to reduce MAPE and NRMSE by 7% and 9%

respectively. A feature importance analysis using SHAP showed that calendar variables, namely week and season, had the greatest importance for the model. Temperature, notably minimum and maximum daily temperature, also had a large influence on model output.

Moon et al. [9] utilized ensemble learning and XAI methods for short-term residential building electricity consumption forecasting. They performed an extensive comparison between five ensemble learning models and ten deep learning models. It was found that the decision tree-based ensemble methods were superior in handling diverse and noisy data, rendering them useful for electricity consumption forecasting. They highlight robustness, interpretability, and efficiency as advantageous characteristics in ensemble models. They argue that ensemble models can handle data that is not extensively preprocessed and that they require less data than deep learning models. A feature importance and SHAP analysis was performed on several ensemble models, revealing that temperature-humidity index and wind chill temperature had significant importance for the forecasting. The SHAP analysis combined with the decision tree-based models provided interpretive insights about complex interactions in the data.

XAI approaches have also been applied to long-term forecasting [32]. Chakraborty et al. [33] predicted long-term cooling energy consumption for residential and commercial buildings under different climate change scenarios. They used SHAP to explain the results and concluded that the XAI method increased the interpretability of the XGBoost prediction model. Another long-term study was conducted by Wenninger et al. [34], who predicted the annual energy consumption for German households, and proposed a novel QLattice model for enhanced explainability. They found that the QLattice model was appropriate for the task of energy consumption forecasting, even though XGBoost achieved slightly better predictive performance. However, a significant strength with the QLattice algorithm is its transparent design, which was found to greatly increase the model explainability compared to established models such as ANN, XGBoost, and Support Vector Regression (SVR).

[With this, it is clear that there is a gap in research on XAI for medium-term forecasting. And in general, the consensus is that more research in the XAI field in the energy sector is needed (?)]

4.4 Residential Electricity Consumption Forecasting

[Forecasting for residential buildings is special and underexplored, but important] [25]

[35]

5 Methodology

[Should change entire method to past tense]

This chapter describes the experimental methodology applied to answer the research questions. The experiments aim to develop monthly electricity consumption forecasts for residential households, with a focus on forecast explainability. A variety of input variables, including historical electricity consumption, household information, weather variables, and calendar information, are used to provide diverse information to the forecasting models. Five decision tree-based ensemble learning models are utilized for the forecasting, and their predictive performance as well as their explainability is assessed and compared. The explainable AI method *SHAP* is used to investigate the effect and importance of the model features. Figure 1 shows the general architecture of building the explainable electricity consumption forecasts.

5.1 Data Preparation and Analysis

This study utilizes electricity consumption data from Umeå Energi, along with external variables in the form of weather data and holiday information. This section describes the data, presents analytical insights as part of enhancing the explainability of the forecasts, and details how the data is preprocessed before performing forecasts.

5.1.1 Electricity Consumption Data

The Umeå Energi dataset contains monthly electricity consumption from 14 286 Swedish households, consisting of 4 years of measurements retrieved from January 1st 2022 to November 30th 2025. This data is provided by Swedish electricity company Umeå Energi. For the purpose of this research, households are defined as unique customer and facility combinations (i.e., one customer may have several facilities, which are regarded as separate households). Households that have any undefined or less than 10 kWh consumption measurements during the time period are removed from the dataset, leaving 13 692 households with complete measurements. All customers are private customers located in Umeå in the northern part of Sweden, meaning the data represents household electricity consumption in a cold climate. The dataset is summarized in Table 1, and Figure 2 shows the distribution of the electricity consumption for all households during the entire time period.

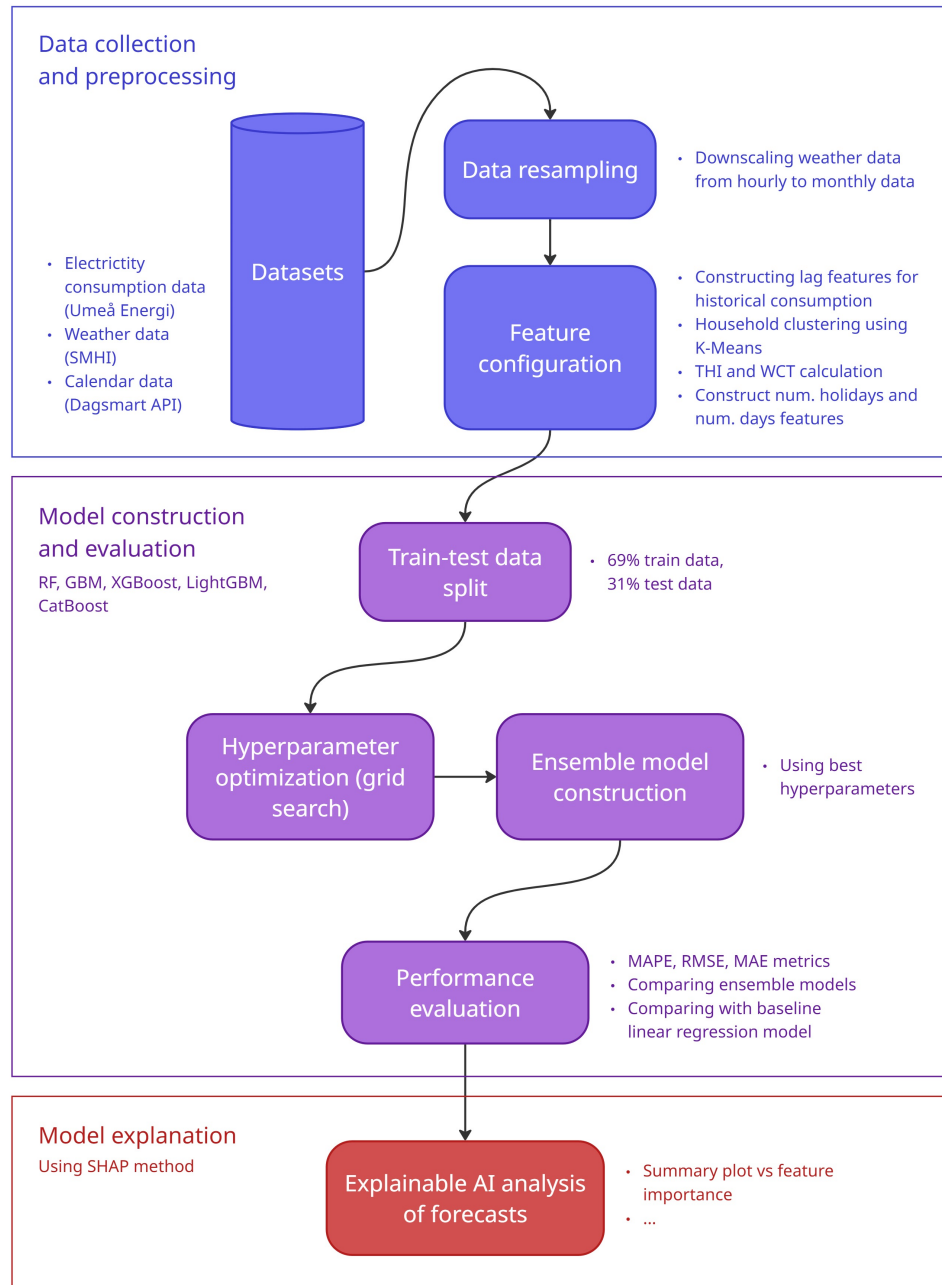


Figure 1: Architecture of modelling explainable electricity consumption forecasts [fix this figure].

Table 1 Summary of the Umeå Energi electricity consumption data.		
Statistic	Umeå Energi Data (kWh)	
Minimum	10	
First quartile	124	
Median	220	
Mean	465	
Third quartile	489	
Maximum	13 490	
Time range	2022-01-01 to 2025-11-30	
Location	Umeå, Sweden	
Number of households	13 692	

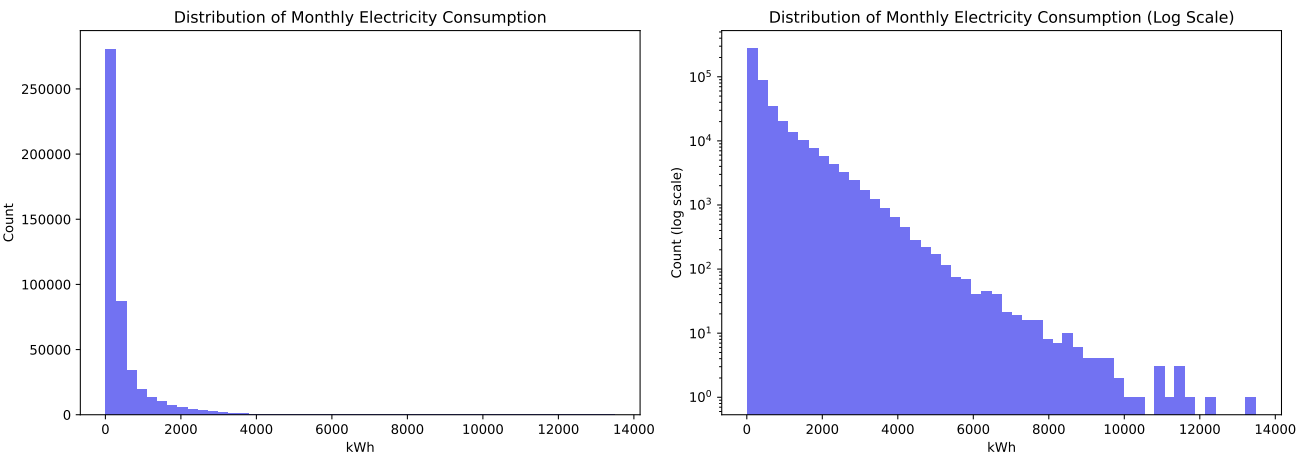


Figure 2: Histogram over the monthly electricity consumption distribution from January 2022 to November 2025.

Figure 2 and Table 1 show that a vast majority of households consume electricity in a similar low range per month, but that there are outliers where the monthly consumption exceeds several thousand kWh. Figure 3 shows the average monthly consumption for all households. It is clear that there is a significant seasonal pattern in the consumption, with more electricity being consumed in the winter and less in the summer. This makes sense for the cold climate, since the winter requires more heating for households with electrical heating, while summers are cool enough to not require much electrical cooling. A warmer climate may have the opposite consumption pattern, since warm summers will have a higher need for cooling measures [33, 36]. Figure 4 illustrates the average consumption by month, showing that consumption varies more during the winter months, while the average consumption is more stable during the summer.

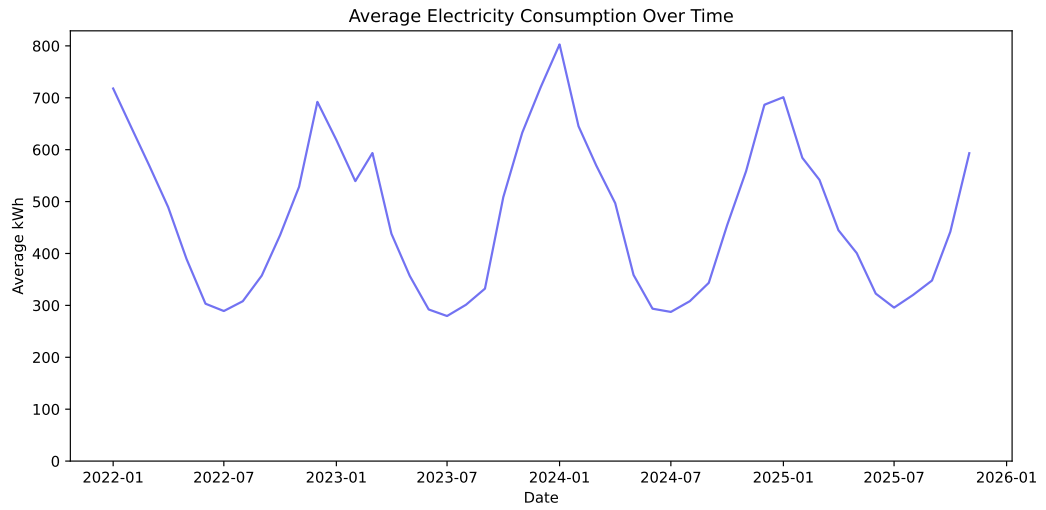


Figure 3: Average household electricity consumption from January 2022 to November 2025.

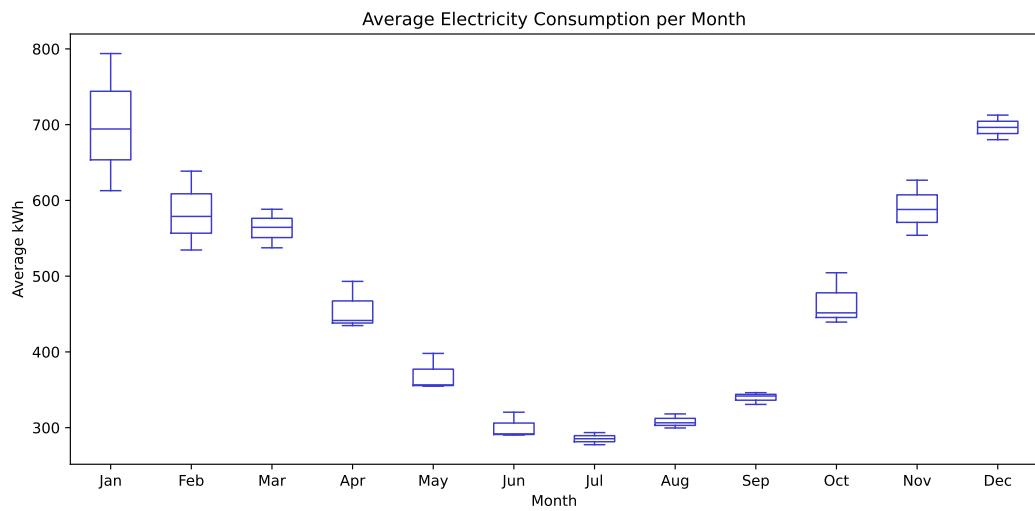


Figure 4: Average household electricity consumption by month from January 2022 to November 2025.

Three input features are constructed from the historical electricity consumption; *lag_1* which represents the consumption of the previous month for the target household, *lag_12* which represents the consumption for the same month the previous year for the target household, and *rolling_mean_3* which represents the average consumption over the past three months for the target household. These variables express the recent consumption patterns, but also the monthly dynamic. Since the year 2022 does not have any past year historical consumption to refer to, it is removed from the training data.

Household Information

In addition to the historical electricity consumption, the Umeå Energi dataset contains information about the households. Households are labeled by type of residence, type of heating system, and type of contract agreement with Umeå Energi. The type of residence is either house, apartment, or vacation home. Heating can be electrical, non-electrical, or district heating. Contract agreements may be variable, fixed, or default assigned contracts. The share of households in each category is found in Table 2. Any given household may change between categories over time (e.g., change its type of heating system), and therefore, the household information is updated per month. Subsequently, Table 2 is not presented in a number of households per category, but rather, the share of observations that occupy a given category.

These features are not directly maintained by Umeå Energi, but instead, the labeling is done by Umeå Energi based on certain properties of the customer, which may cause the labeling to be inaccurate in some cases. However, it provides a good estimation of the household properties.

Table 2 Household types in the Umeå Energi dataset.

Type	Percentage
Apartment	58%
House	38%
Vacation home	4%
Non-electrical heating	70%
Electrical heating	19%
District heating	11%
Variable contract	82%
Default contract	18%
Fixed contract	< 1%

Household Clustering Using K-Means

Since there are many households in the electricity consumption data, it is relevant to not only group households by their properties, but also by their consumption patterns. The idea is to identify groups where the similarity within the group is higher than that among the groups [29]. K-Means clustering is one of the most popular algorithms for solving this problem in a simple way. K-Means is an unsupervised learning algorithm, where the main idea is to select k centroids, one for each cluster, and place them as far from each other as possible. The next step is to associate each point in the dataset with the closest centroid. After associating all points, the k centroids are recalculated by averaging the points within each cluster. This process is repeated until the centroids no longer move. Formally, the algorithm aims at minimizing the squared error function $W(S, C)$ defined in Equation 5.1 [37].

$$W(S, C) = \sum_{k=1}^K \sum_{i \in S_k} \|y_i - c_k\|^2 \quad (5.1)$$

In the equation, S is a partition of k clusters represented by observations in the form of vectors y_i . Each cluster S_k is non-empty and non-overlapping, and has a centroid c_k ($k = 1, 2, \dots, K$).

In these experiments, K-Means clustering is applied to define household groups based on their electricity consumption. The elbow method is used to find an optimal number of k , i.e. the number of clusters. The elbow method is a visual method, where different values of k are plotted against the variance within the clusters [38]. As the value of k increases, the average variance becomes smaller. The goal is to find a value of k for which the variance stops decreasing sharply, that is, the elbow of the graph. Figure 5 shows the elbow plot for determining the optimal number of household groups, in this case $k = 3$. This results in three household groups; low consumption, medium consumption, and high consumption. Each household in the dataset is assigned a label based on the cluster it belongs to, and this label becomes an input feature for the forecasting models. Table 3 details each cluster regarding size and consumption patterns. The large standard deviation present in each cluster suggests that outlier months with high consumption exist for all identified consumption patterns.

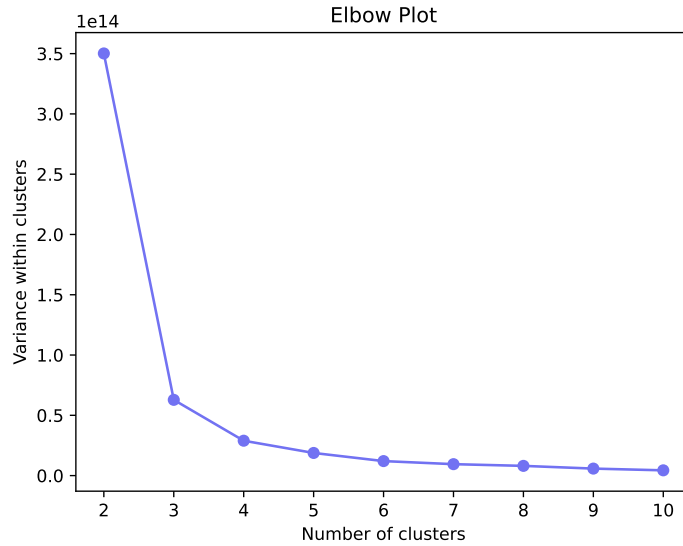


Figure 5: Elbow plot for finding the optimal number of household consumption profile clusters k .

Table 3 Clustered consumption profiles in the Umeå Energi dataset.

Cluster	Households	Mean (kWh)	Median (kWh)	SD (kWh)
Low consumption	1 809	369	174	530
Medium consumption	4 559	419	232	556
High consumption	7 324	528	233	730

Variable Interactions

Interactions between the variables are investigated using a correlation map, as shown in Figure 6. This correlation map includes the internal historical consumption features as

well as external features in the form of weather indices (THI, WCT see Equation 5.2 and Equation 5.3) and calendar information (number of holidays, number of days). The three historical consumption variables show a high correlation with each other, while they have a low correlation with all the external variables, indicating that the external variables may contribute information to the forecasts that internal features are not able to capture.

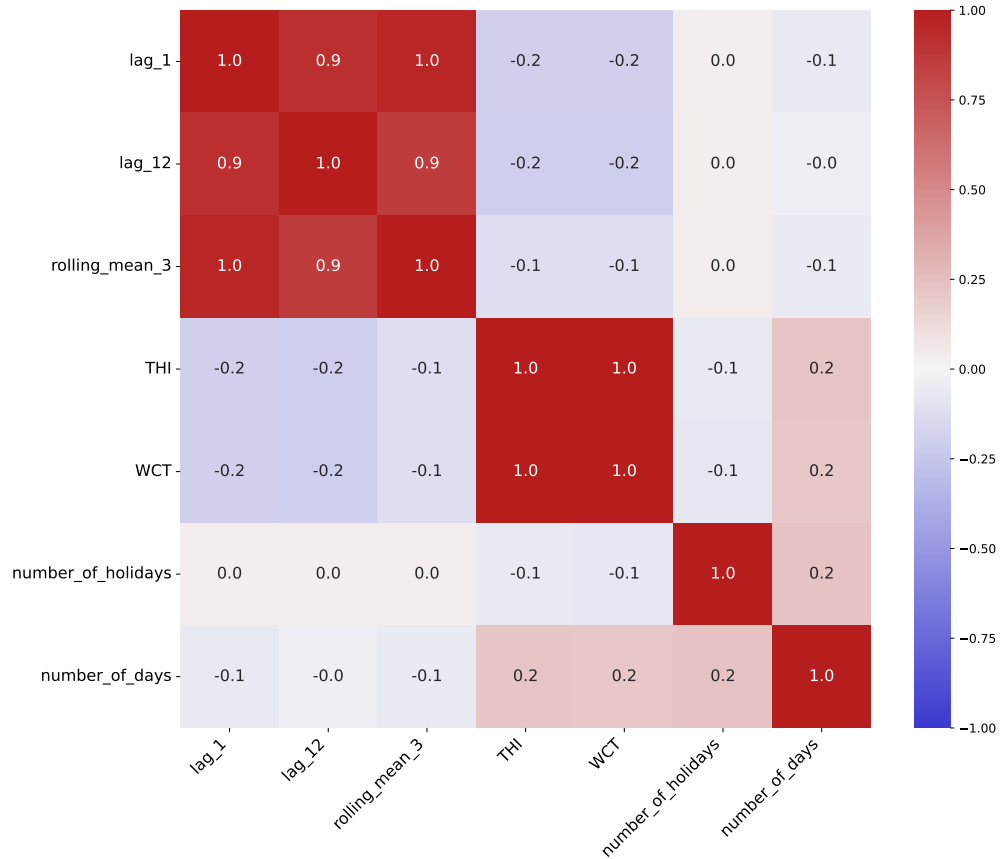


Figure 6: Correlation map of internal variables, weather indices, and holiday indicators.

5.1.2 External Variables

Along with the historical electricity consumption data from Umeå Energi, weather variables are retrieved from SMHI¹ and calendar information is retrieved from Dagsmart API². These external variables are intended to improve the forecasts by including environmental factors that may affect consumption patterns.

Weather Data

Weather data is retrieved from SMHI's open data for the January 1st 2022 to November 30th 2025 time period. Nine variables are selected from the Umeå Airport measurement

¹Swedish Meteorological and Hydrological Institute, SMHI. <https://www.smhi.se/data> (Accessed 2026-03-17)

²Swedish calendar data API, Dagsmart API. <https://dagsmart.se/api/> (Accessed 2026-03-16)

station, while three variables pertaining to solar attributes are retrieved from the Umeå Sol station, and snow depth is retrieved from Umeå Röbbäcksdalen. All variables are obtained as hourly measurements, except snow depth which is retrieved at daily intervals. Table 4 details the weather variables and their measurement units. Across all variables, 4.9% of the measurements are missing, where most missing values occur in the lowest cloud base variable that has 42% missing values.

The measurements of all variables are aggregated to monthly values using mean. Figure 7 shows the monthly values of each weather variable plotted over the entire time range. The seasonal component is apparent for many of the weather variables, while some variables such as wind speed and air pressure lack a clear seasonal pattern. This creates potential for the different weather variables to contribute to the forecasts in unique ways.

Table 4 Weather variables retrieved from SMHI.

Variable	Description	Resolution
<i>temperature</i>	Air temperature (°C)	Hourly
<i>dew_point_temp</i>	Dew point temperature (°C)	Hourly
<i>snow_depth</i>	Snow depth (m)	Daily
<i>humidity</i>	Relative humidity (%)	Hourly
<i>wind_direction</i>	Wind direction (°C)	Hourly
<i>wind_speed</i>	Wind speed (m/s)	Hourly
<i>lowest_cloud</i>	Lowest cloud base (m)	Hourly
<i>global_radiation</i>	Global irradiance (Sweden) (W/m ²)	Hourly
<i>longwave_radiation</i>	Long wave irradiance (W/m ²)	Hourly
<i>sunshine</i>	Sunshine amount (s)	Hourly
<i>air_pressure</i>	Air pressure (hPa)	Hourly
<i>visibility</i>	Visibility (m)	Hourly

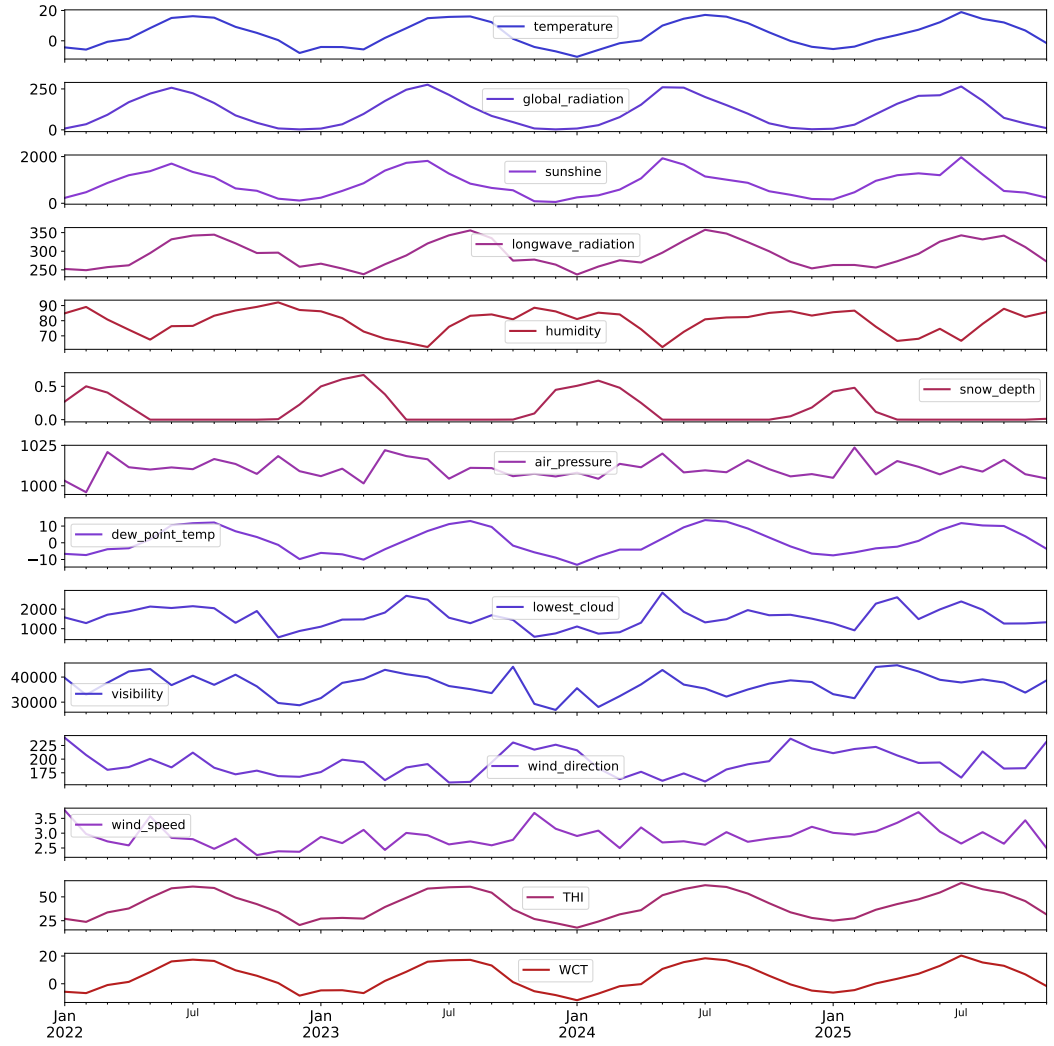


Figure 7: Weather variables' monthly values between January 2022 and November 2025.

Although weather factors have been proven to be useful for predicting electricity consumption, the link between weather and electricity consumption can be indistinct. Weather indices incorporating temperature, humidity, and wind speed can provide a better representation of the perceived environmental effects and the impact on electricity consumption [9]. Two indices, temperature humidity index (THI) and wind chill temperature (WCT), are calculated as in Equation 5.2 and Equation 5.3. In the equations, *temp* is temperature, *humi* is humidity, and *ws* is wind speed. These indices are included as input features for the forecasting models.

$$THI = (1.8 \times temp + 32) - [(0.55 - 0.0055 \times humi) \times (1.8 \times temp - 26)] \quad (5.2)$$

$$WCT = 13.12 + 0.6215 \times temp - 11.37 \times ws^{0.16} + 0.3965 \times temp \times ws^{0.16} \quad (5.3)$$

Calendar Information

In addition to weather data, calendar information is included as external factors. During holidays, energy consumption behaviors change due to differences in social patterns. Energy consumption is not only dependent on the decision-making of individuals, it also relies on the cultural context of social and economic activities [39]. Holiday information is retrieved from Dagsmart, an open Swedish calendar information API. All holidays and weekend days during the years 2022 through 2025 are obtained, and these are compiled to a *number of holidays* feature that indicates the number of holidays for each month in the dataset. In Figure 8, the number of holidays per month is visualized, showing that December is the month with the most holidays. The fewest holidays are during the rest of the winter and autumn months. An additional *number of days* feature is constructed for each month in the dataset, since the number of days varies (e.g., February is missing several days of consumption compared to months with 31 days).

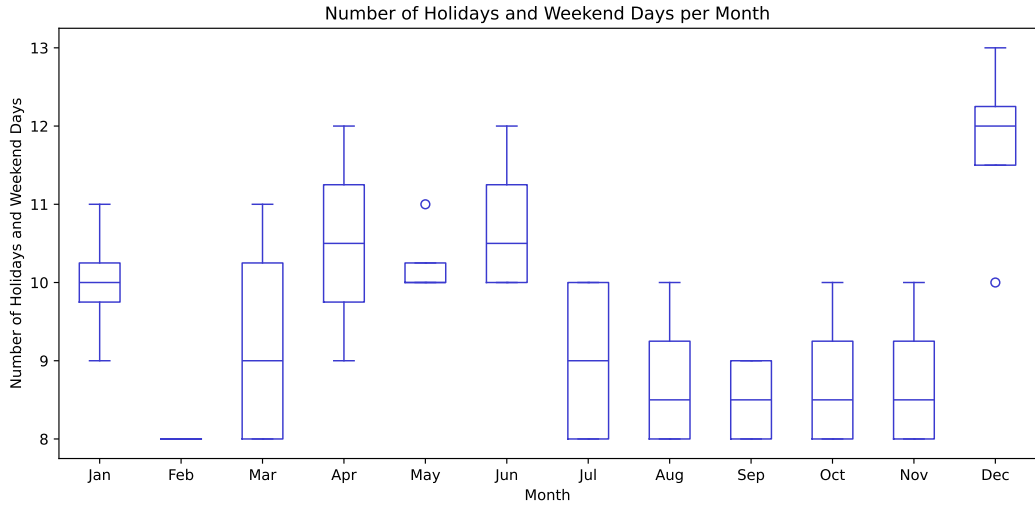


Figure 8: Number of Swedish holidays per month during the years 2022 through 2025.

5.1.3 Feature Selection

The historical consumption, weather, and calendar variables are compiled to a complete feature set for the forecasting models. A feature selection is made from the weather variables, based on the correlation maps found in Figure 20 and Figure 6. Variables with a high correlation to several other variables are excluded from the feature set, in this case global radiation, long wave radiation, dew point temperature, and visibility. This is due to highly correlated features accounting for each other's impact to the model output [formulera bättre], and the SHAP method assuming that features are uncorrelated [23]. The complete selected feature set is found in Table 5. A subset of 2 000 random households are selected for the model training and evaluation, representing 15% of the total number of households. The data is then split into training and test data such that the years 2023 and 2024 are the training set, and January through November 2025 is the test data (see Table 6).

Table 5 Selected input features for the forecasting models.

Feature	Description	Data Type
<i>month</i>	Month of the year	Numeric
<i>year</i>	Year	Numeric
<i>number_of_holidays</i>	Number of holidays and weekend days during month	Numeric
<i>number_of_days</i>	Number of days during month	Numeric
<i>cluster</i>	Household cluster	Household information (categorical)
<i>building_type</i>	Type of building	Household information (categorical)
<i>heating_type</i>	Type of heating system	Household information (categorical)
<i>contract_type</i>	Type of contract	Household information (categorical)
<i>temperature</i>	Monthly average air temperature (°C)	Weather condition (numeric)
<i>snow_depth</i>	Monthly average snow depth (m)	Weather condition (numeric)
<i>humidity</i>	Monthly average relative humidity (%)	Weather condition (numeric)
<i>wind_direction</i>	Monthly average wind direction (°C)	Weather condition (numeric)
<i>wind_speed</i>	Monthly average wind speed (m/s)	Weather condition (numeric)
<i>lowest_cloud</i>	Monthly average lowest cloud base (m)	Weather condition (numeric)
<i>sunshine</i>	Monthly average sunshine amount (s)	Weather condition (numeric)
<i>air_pressure</i>	Monthly average air pressure (hPa)	Weather condition (numeric)
<i>THI</i>	Temperature humidity index	Weather index (numeric)
<i>WCT</i>	Wind chill temperature index	Weather index (numeric)
<i>lag_1</i>	Consumption for the previous month (kWh)	Historical consumption (numeric)
<i>lag_12</i>	Consumption for the same month last year (kWh)	Historical consumption (numeric)
<i>rolling_mean_3</i>	Average consumption for the previous three months (kWh)	Historical consumption (numeric)

Categorical features in the dataset are label encoded to integer values for all models except CatBoost. This means that each categorical value is represented by a unique integer, instead of its original string value. Label encoding, or ordinal encoding, is one of the simplest encoding techniques, and it may introduce an ordering among categorical variable values, even if one does not exist. However, for decision tree models in particular, there can be a favorable trade-off between performance and time cost by selecting label encoding [40]. For CatBoost, the original string values are preserved due to the fact that CatBoost is specifically designed to handle categorical features [41].

Table 6 Train-test split of the dataset.

Data	Time range	Percentage	Data points
Training data	January 2023 to December 2024	69%	48 000
Test data	January 2025 to November 2025	31%	22 000

5.2 Ensemble Learning Forecasting Models

Forecasting is performed using different decision tree-based ensemble learning models. Decision trees classify data items by asking a sequence of questions about the items' features. The nodes of the tree contain questions, and for each possible answer, every internal node points to one child node. The sorting of an item into a class begins from the root of the tree, and according to the item's answer to the questions, it follows the path to a leaf node [42]. An example of a decision tree structure is found in Figure 9. One large advantage with decision trees is that they are considerably easier to interpret than neural networks, as they follow logical rules instead of having connections with numeric weights between nodes. Decision trees perform well for a few highly relevant attributes, but not as well if there are many attributes with complex interactions. Ensembles of decision trees can be employed to increase accuracy when the training data is more complex and changing [43]. Decision tree ensembles are considered among the most reliable and precise methods for regression and classification tasks, and they have been shown to be effective for predicting energy consumption [9]. However, one disadvantage is that the interpretability decreases when multiple classifiers are involved, making it more difficult to understand the reasoning behind the ensemble's decision [43]. This creates a need for model explainability outside the ensemble's inner workings, which in this thesis is addressed by applying explainable AI analysis.

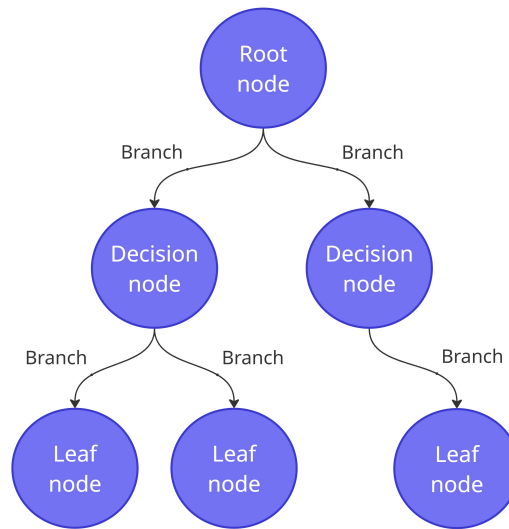


Figure 9: A decision tree.

Five decision tree-based ensemble models are selected as forecasting models; random forest (RF), gradient boosting machine (GBM), extreme gradient boosting machine (XGBoost), light gradient boosting machine (LightGBM), and categorical boosting (CatBoost). These models are selected due to their reliability, precision, and robustness in regression and classification tasks, and their ability to capture nonlinear relationships [9].

Random forest is a decision tree ensemble learning algorithm where the trees depend on a set of random feature splits at each node. This means that every tree produces its own prediction according to its set of data. The individual predictions are then combined into a final prediction via majority voting. RF is an extension of the bagging method, efficient for large datasets and robust to outliers and overfitting, making it suitable for this application.

Additionally, RF is faster than other bagging or boosting techniques [44].

Gradient boosting machines are a family of ML algorithms where the learning process consecutively fits a new tree with the goal of reducing the errors of the previous one. This is done by constructing new base-learners that are maximally correlated to a decreasing loss function. The loss function can be of various kinds, which makes GBM highly flexible for different data-driven tasks. It is up to the model designer to choose the most appropriate loss function [45]. GBM is able to learn complex patterns and interactions in the data due to its iterative learning process [9].

XGBoost is a specialization of the gradient boosting algorithm with several optimizations, creating a scalable tree boosting system. The optimizations include a sparsity-aware algorithm for handling missing values, approximate learning via a weighted quantile sketch algorithm, and efficient cache access patterns reducing runtime. This results in a fast and robust boosting model using minimal resources [46].

LightGBM is another gradient boosting algorithm that focuses on efficiency for cases with many input features and when the amount of data is large. LightGBM uses two novel techniques to improve performance and efficiency, *gradient-based one-side sampling* (GOSS) and *exclusive feature bundling* (EFB). GOSS excludes data points with smaller gradients that contain less information, with the goal filtering out noisy data while retaining information gain. EFB bundles mutually exclusive features (that rarely have nonzero values at the same time) to create a smaller feature space with nearly exclusive features [47].

CatBoost is a gradient boosting algorithm that uses a technique called *ordered boosting* to reduce prediction shift caused by target leakage which occurs in other gradient boosting algorithms. CatBoost also encodes categorical features uniquely, by applying a technique called *ordered target statistics* to handle high cardinality features [41]. Because of this, CatBoost is able to achieve high accuracy, especially when the data contains a significant number of categorical features [9].

5.2.1 Configuration of Models

To configure the models, a search for the best hyperparameters is performed via exhaustive grid search. The optimization consists of a five-fold time series cross-validation process over the hyperparameters specified in Table 7, using the scikit-learn packages GridSearchCV and TimeSeriesSplit. A subset of 30% of the training dataset is used for the grid search. For XGBoost, the grid search is minimized to 50 random combinations of the search space, in order to reduce execution time.

Table 7 Hyperparameters for decision tree ensemble models.

Model	Hyperparameters
RF	Trees count: 128 Features per split: auto, sqrt, log2
GBM	Iteration count: 100, 250, 500 Learning rate: 0.01, 0.05, 0.1 Depth: 5, 10 Loss type: quantile, Huber
XGBoost	Iterations: 250, 500, 1000 Learning rate: 0.01, 0.05, 0.1 Depth: 6, 8, 10 Subsampling rate: 0.5, 0.75, 1.0 Feature sample by tree/level/node: 0.5, 0.75, 1.0 Booster type: gbdt, dart
LightGBM	Iterations: 1000, 1500 Learning rate: 0.01, 0.05, 0.1 Leaves: 64 Subsample: 0.5 Feature sample by tree: 1.0 Booster type: gbdt, dart
CatBoost	Learning rate: 0.03, 0.1 Max tree depth: 4, 6, 10 L2 regularization levels: 1, 3, 5, 7, 9

As a baseline comparison, an existing linear regression forecasting model developed by Umeå Energi is utilized. The model is based on historical electricity consumption and outdoor temperature.

5.3 Evaluation Metrics

To evaluate the predictive performance of the forecasting models, three statistical metrics are used; mean absolute percentage error (MAPE), root mean squared error (RMSE), and mean absolute error (MAE). MAPE is one of the most common metrics to evaluate forecasting accuracy, due to its scale-independence and interpretability. It defines the average of the absolute percentage errors, as in Equation 5.4. One disadvantage to be aware of is that MAPE yields infinite or undefined values when the actual values are zero or close to zero [48]. This experiment handles this by removing any households from the dataset that have less than 10 kWh consumption during some month. RMSE and MAE, as defined in Equation 5.5 and Equation 5.6, are utilized to give insights on the size of the errors in kWh. These two metrics are also among the most commonly used in the electricity consumption forecasting field [25].

$$MAPE = \frac{1}{n} \times \sum \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100 \quad (5.4)$$

$$RMSE = \sqrt{\frac{1}{n} \times \sum (y_t - \hat{y}_t)^2} \quad (5.5)$$

$$MAE = \frac{1}{n} \times \sum |y_t - \hat{y}_t| \quad (5.6)$$

In the equations, y_t denotes the actual observed value at time t , while $\hat{y}_t$ denotes the predicted value at time t . The total number of prediction instances is denoted by n .

5.4 Explainable AI Analysis

To enhance the interpretability of the forecasts and investigate the influence of model features, a SHAP analysis is performed for all models. The SHAP methodology consists of calculating the Shapley values for each predictor. Individual predictions are evaluated and then aggregated to create a global interpretation. This experiment uses the SHAP *TreeExplainer*, specifically designed for the structure decision tree models, where the explainer traces the decision paths formed by the series of decision splits made by the models [24]. For each ensemble model, both the models' built-in feature importances and the calculated SHAP values are obtained. The results are then visualized via a feature importance plot, and a SHAP summary (beeswarm) plot displaying both feature importance ranking and the impact of individual feature values on model output.

5.5 Experimental Setup

Experiments are performed in the Python language. To process and analyze the data, the *pandas*, *seaborn*, and *matplotlib* packages are utilized. The *scikit-learn* package provides the RF and GBM models, as well as *GridSearchCV* for the exhaustive grid search. XGBoost, LightGBM, and CatBoost models are obtained from their respective GitHub repositories. The *shap* Python package is used for the explainable AI analysis.

Experiments are run in a Windows 11 virtual environment with two Intel Xeon Gold 6526Y CPU's and an Nvidia A16-2Q 2GB GPU. For compatible models, GPU acceleration is used when performing grid search, training the models, and performing the SHAP analysis.

6 Results

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This chapter presents the results to the experiments performed aiming to answer the research questions. Firstly, the performance of the forecasting models is presented, as well as a comparison with the baseline model. Secondly, results of the explainable AI analysis is presented.

6.1 Performance of Forecasting Models

The performance of the five decision tree-based ensemble models was assessed via MAPE, RMSE, and MAE metrics. Table 8 shows the scores for each model, separated by internal features and total features. Scores under the internal feature category represent the models' performance when trained solely using the historical electricity consumption, while total features refers to the combination of internal and external features. The best performing model was XGBoost, with 14.96% errors on the total feature set. RF showed the worst performance with 16.33% errors and considerably higher RMSE and MAE scores on the total feature set, compared to the other models. Looking at the difference between inclusion and exclusion of external features, adding the external features results in a higher forecast accuracy in terms of MAPE. On the contrary, MAE and RMSE scores worsen for the total feature group in almost all cases.

Table 8 Performance of forecasting models.						
Model	Internal features			Total features		
	MAPE (%)	RMSE (kWh)	MAE (kWh)	MAPE (%)	RMSE (kWh)	MAE (kWh)
RF	16.34	142.47	65.93	16.33	171.47	74.68
GBM	15.67	144.07	64.49	15.07	154.97	66.67
XGBoost	15.40	144.12	63.94	14.96	153.55	65.46
LightGBM	15.67	150.21	66.09	15.18	159.54	67.03
CatBoost	15.73	152.85	65.74	16.05	150.07	66.52

A pattern occurring for all models is that on average, the models overestimate the electricity consumption during the winter months in the beginning of the year, as illustrated in Figure 10. During the summer, the average predictions lie closer to the actual average consumption of the households. The scatter plots in Figure 11 show that the overestimation of electricity consumption is a general trend for all models except XGBoost, indicated by the regression lines with larger slopes than the perfect prediction line. For those models, it is clear that predictions in the several thousand kWh range are commonly overestimated,

while the errors for lower kWh predictions are more evenly distributed around the perfect prediction line.

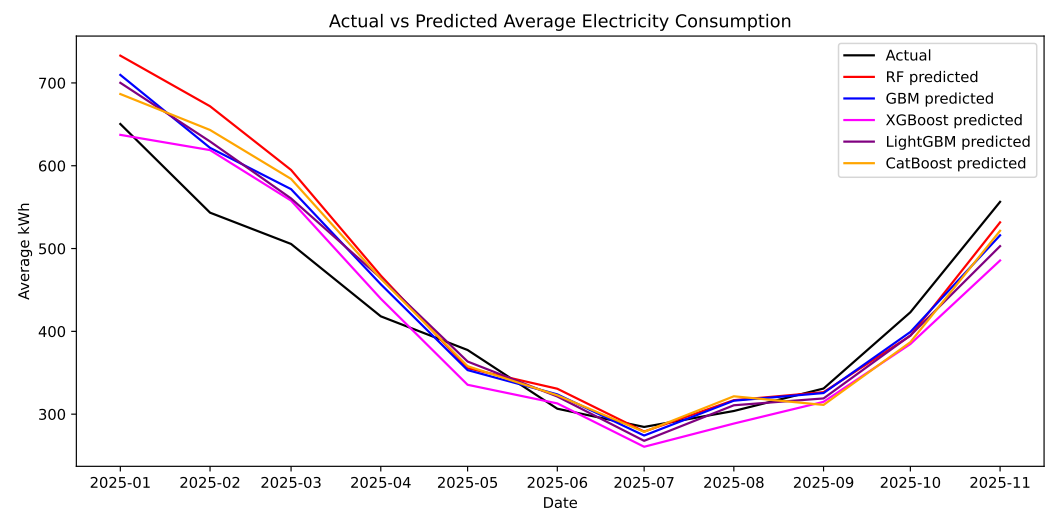


Figure 10: Average actual versus predicted electricity consumption for all models.

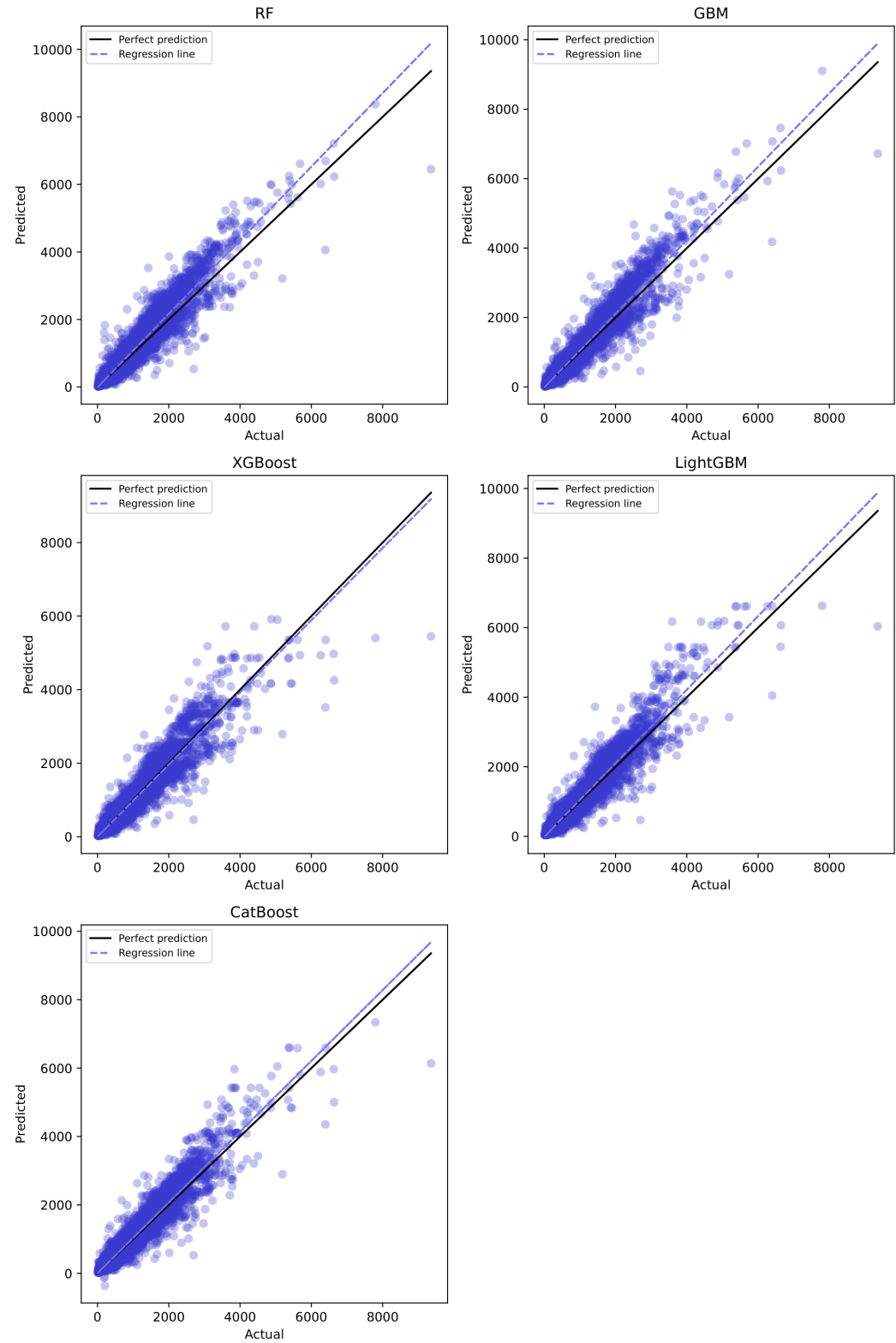


Figure 11: Scatter plots showing prediction error distribution for each model.

6.1.1 Comparison with Baseline Model

The baseline linear regression model used for comparison utilized historical consumption and temperature as input variables. The past 24 months of electricity consumption and monthly average temperature was used to predict the following month's consumption. Evaluation of the linear regression model was performed using the same 2000 household sample as for the ensemble models, and a resulting comparison between the linear regression model and the best ensemble is found in Table 9. The linear regression model performs worse by roughly 3.5% in percentage errors. RMSE and MAE scores are also worse for the linear model, especially when compared to ensemble models using only internal features.

Table 9 Performance of best ensemble model versus linear regression.

Model	MAPE (%)	RMSE (kWh)	MAE (kWh)
Best ensemble (XGBoost)	14.96	153.55	65.46
Linear regression	18.47	161.43	71.98

Figure 12 shows the average predicted monthly electricity consumption for the linear regression model and all ensemble models. The linear regression model shows the same tendency to overestimate the consumption during the beginning of the year as the ensemble models do, while also showing a pattern of underestimating the consumption during the summer.

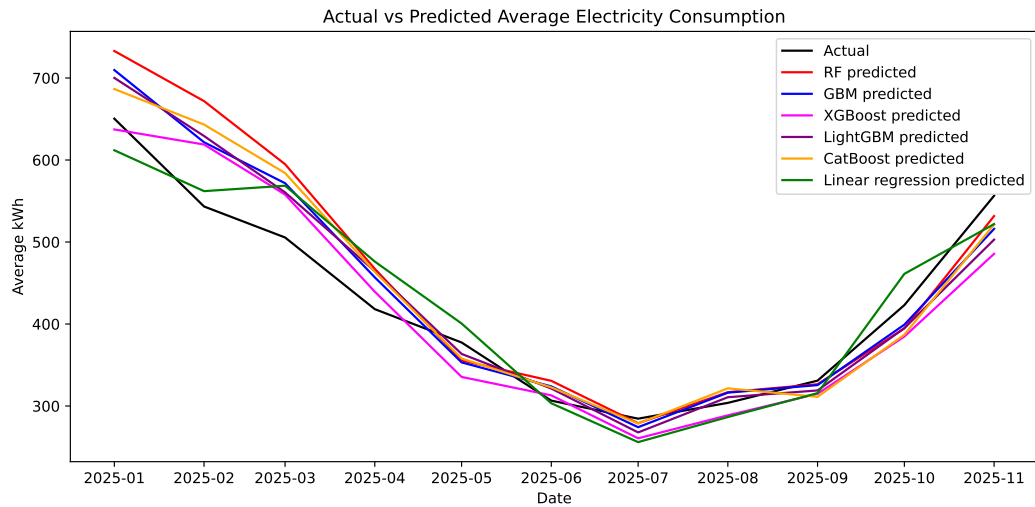


Figure 12: Average actual versus predicted electricity consumption for ensemble models and linear regression.

6.2 Explainable AI Analysis of Forecasts

After assessing the predictive performance of all five ensemble learning models, the models' predictions were analyzed via the SHAP method. Firstly, the models were trained using their optimal hyperparameters, and feature importance metrics were retrieved for the trained models. Secondly, a tree SHAP analysis was performed to calculate the Shapley values

for all features. For the RF model, approximate SHAP values¹ were used due to the full calculation being time-consuming for this particular model. Figures 13 to 17 show the resulting feature importance plots (left) and SHAP summary plots (right) [change to a) and b)] for each model.

Both the feature importance and SHAP analysis show that the *lag_1* and *lag_12* historical consumption features are highly influential for all models. According to the feature importance plot for GBM in Figure 14, the model almost exclusively relies on the *lag_1* and *lag_12* features. The SHAP summary plots for all models, showing the effect of individual feature values on the prediction, also demonstrate that these features have a clear relationship where low feature values contribute to a lower predicted value, and high feature values contribute to higher predictions. Most models prioritize the *lag_1* feature over *lag_12* in their decision-making, except for RF. The *rolling_mean_3* feature is ranked relatively high for all models, but the SHAP summary plots show that the feature may have a more complex relationship to the model output for some models. For GBM, XGBoost, and LightGBM, high values of the *rolling_mean_3* feature contribute both to decreasing and increasing predicted values. The XGBoost model deviates from the other models by ranking the *WCT* feature the highest in feature importance. However, the more detailed SHAP summary plot ranks the *lag_1* and *lag_12* as most influential, displaying them with a similar pattern to the other model. The SHAP summary for this model also has a significantly lower ranking of the *WCT* feature.

Household information features specifying residence type, and heating system were influential for the RF model, but remaining models were not as impacted by these features. The contract type feature was ranked low for all models. Even though these features commonly saw low rankings, the SHAP summaries indicate patterns where the features values had consistent impact on the SHAP values. Because of the special categorical feature handling of the CatBoost model, the SHAP summary in Figure 17 is unable to display the high/low feature value indications for these features, but it is evident that some instances of the *contract_type* feature had a large impact on the resulting SHAP value. In addition to the household information features, clustering of the households was performed based on their electricity consumption. The three resulting consumption groups (see Table 3) were encoded by the *cluster* feature. This feature was found in the lower ranks for all models.

Among the external variables, wind direction and sunshine amount were consistently ranked highly for all models. The lowest cloud base feature commonly saw a low ranking, while remaining weather variables had a more varied impact depending on the model. The calculated weather indices *THI* and *WCT* also had a varied impact on the models. This can partly be attributed to the fact that the features are fully correlated with each other and with some of the weather variables [discussion probably]. The two calendar information features, *number_of_holidays* and *number_of_days*, are commonly ranked among the less important features for the models, along with the *year* timestamp variable. The *month* variable, indicating the month of the year, is however ranked highly for several models.

¹SHAP Tree Explainer: https://shap.readthedocs.io/en/latest/generated/shap.TreeExplainer.html#shap.TreeExplainer.shap_values. Retrieved 2026-05-15.

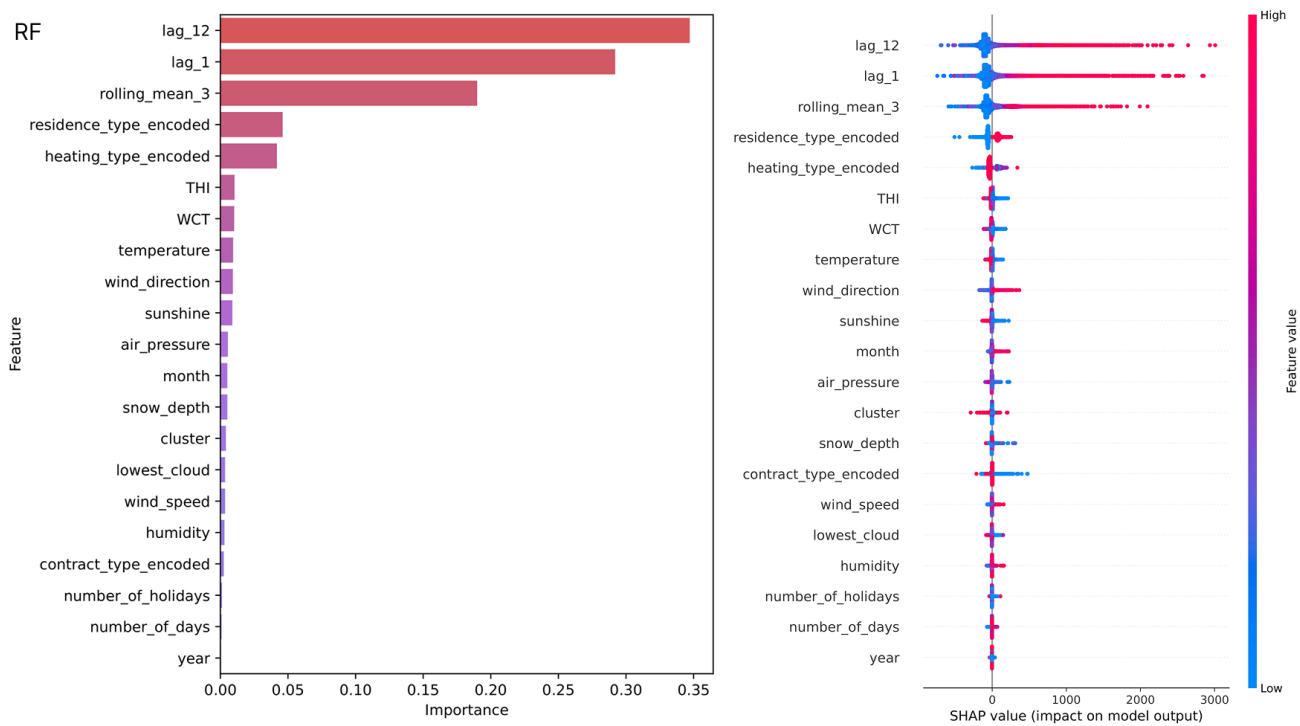


Figure 13: Feature importance and SHAP summary for RF.

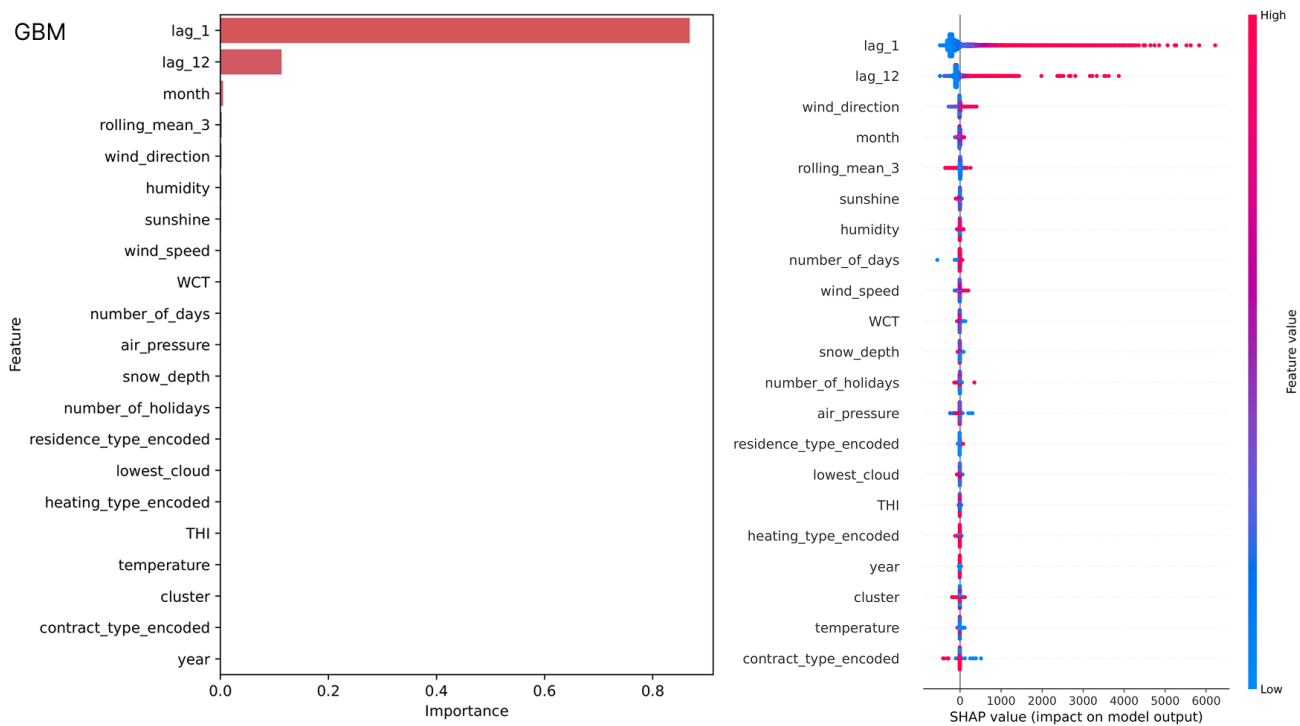


Figure 14: Feature importance and SHAP summary for GBM.

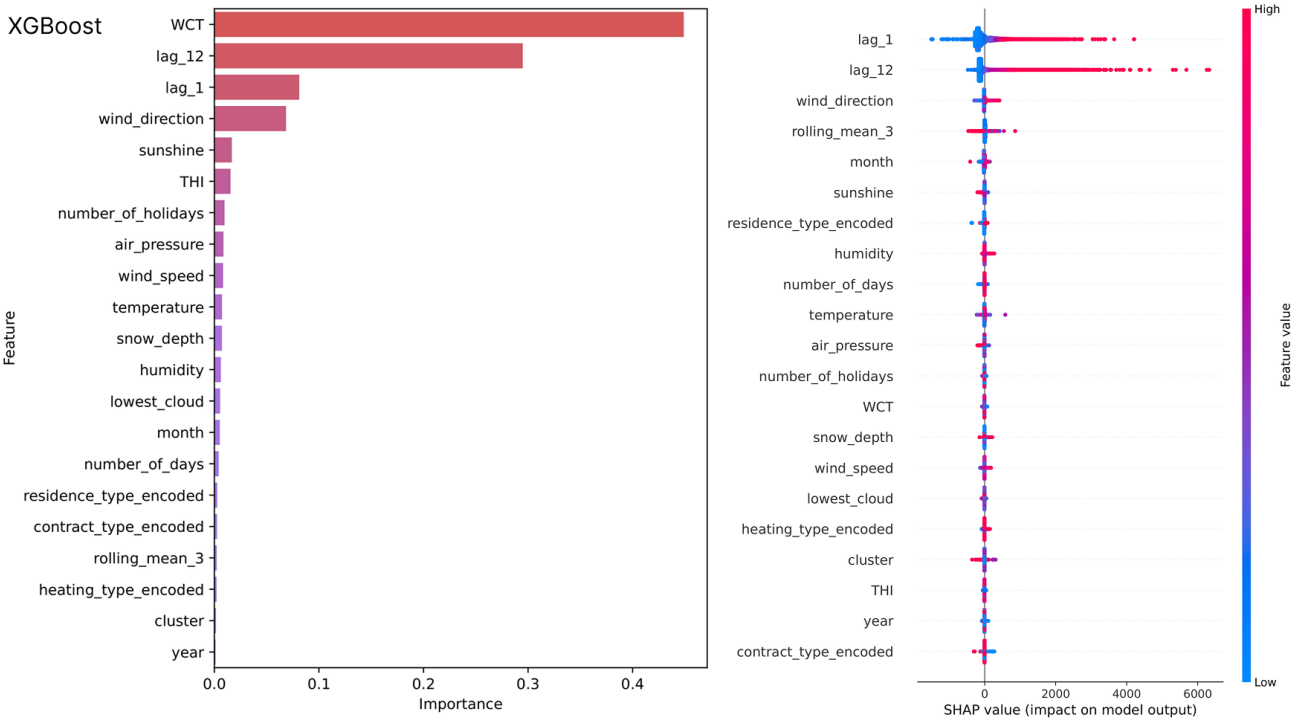


Figure 15: Feature importance and SHAP summary for XGBoost.

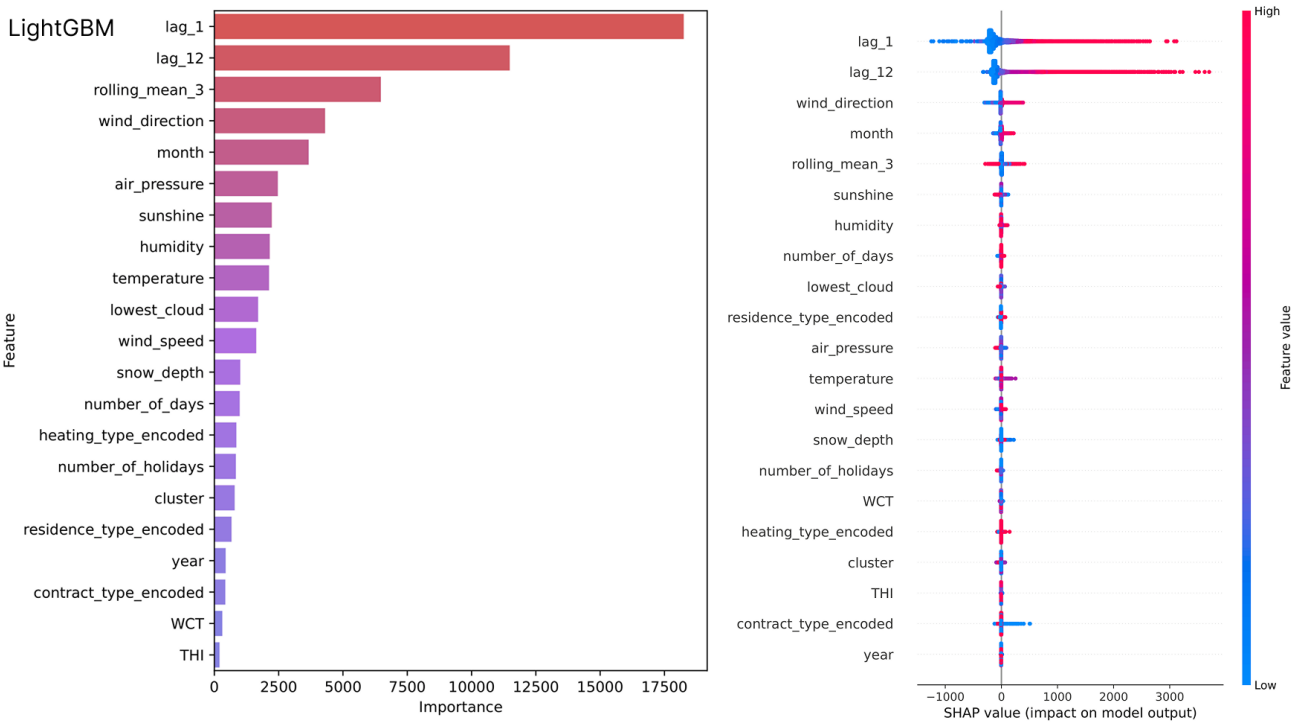


Figure 16: Feature importance and SHAP summary for LightGBM.

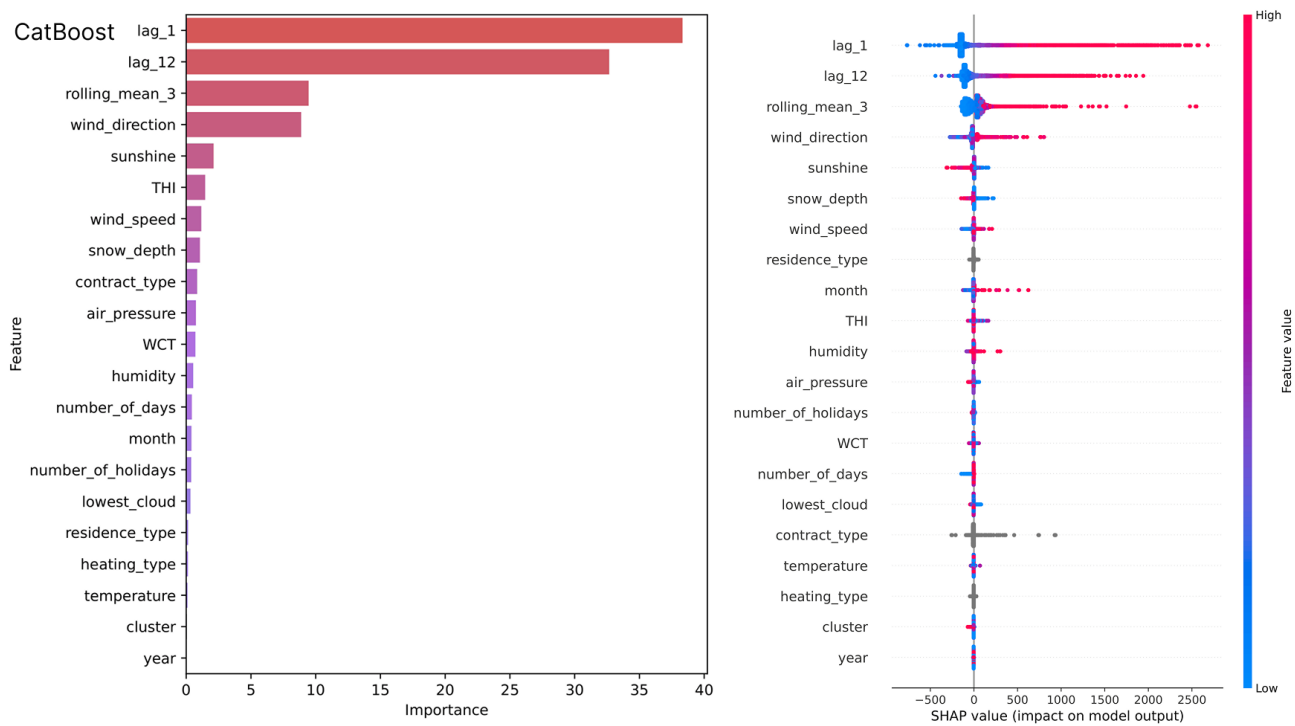


Figure 17: Feature importance and SHAP summary for CatBoost.

7 Discussion

This chapter interprets and discusses the results from chapter 6. Firstly, the predictive performance of the forecasting models is interpreted and further analyzed, and secondly, the contributions of individual features are assessed based the explainable AI analysis. Lastly, methodological limitations of this thesis are discussed, as well as perspectives on the implementation of the forecasting system into a real-world environment.

7.1 Performance of Forecasting Models

The forecast accuracy showed similar results across all ensemble models, with percentage errors around 15%. All models showed a large discrepancy between RMSE and MAE scores in terms of kWh, suggesting that the dataset contains large outliers. Since the RMSE metric squares the errors before averaging, large errors are penalized more, compared to MAE where all errors are treated equally via the absolute value. The considerably higher RMSE scores therefore indicate that there are some outliers with high kWh values, where the error may not be large in percentage, but the error is large in kWh due to the high consumption.

Between the internal and total feature groups, MAPE scores improved when including external features, while errors terms of RMSE and MAE increased for the total feature group. MAPE is more sensitive to errors on small consumption values, while RMSE in particular is sensitive to high consumption values and outliers. This indicates that the external factors help with increasing forecast accuracy for households with a lower and more stable consumption, while households with a high or varied consumption do not benefit from the external factors.

Household energy consumption forecasting is unique because of the stochastic nature of small-scale consumers. When forecasting the aggregated energy consumption for a collection of consumers (e.g., an entire city), the effect of randomness and user behavior is balanced out between consumers. This is not the case when performing forecasts at the individual level, since the individual customer's behavior is highly influential to the consumption [10, 28]. Additionally, this thesis utilizes a dataset with a heterogeneous group of consumers, living in various types of residence and with different kinds of energy demands, which further complicates the forecasting problem.

After obtaining the initial results, further analysis and experiments were performed to investigate the effect of outlier values on the models' performance. The data was modified by introducing a maximum monthly consumption threshold of 2000 kWh. This number was selected due to the majority of observations registering below 2000 kWh, as shown in Figure 2. Households exceeding this threshold (267 households) were removed from the sample of 2000 households used for the experiments. The grid search, training, and evaluation process was repeated for each model with the modified data, and the resulting scores

are found in Table 10.

Model	Internal features			Total features		
	MAPE (%)	RMSE (kWh)	MAE (kWh)	MAPE (%)	RMSE (kWh)	MAE (kWh)
RF	15.99	70.67	38.37	15.32	69.42	37.03
GBM	15.27	70.41	37.29	14.43	66.85	35.11
XGBoost	15.25	70.17	37.74	14.40	68.52	36.27
LightGBM	15.38	72.44	38.52	14.37	66.47	35.30
CatBoost	15.44	70.03	37.34	15.12	66.05	35.53

The removal of high-consuming households resulted in a small improvement in MAPE, and large improvements in both RMSE and MAE, confirming that high-consuming households have a great impact on the models' RMSE and MAE performance. The root mean squared errors were roughly half of the original for all models, and mean absolute errors decreased with around 30 kWh. MAPE scores improved up to 1 percentage point. Notably, the RMSE and MAE scores for the total feature group (both internal and external features) were better than for solely the internal group when the high-consuming households were removed from the data. This is contrary to the previous results, where the models showed better scores only using the historical consumption features. These results suggest that external variables create noise for the households with a very high consumption, and actually worsen the predictive performance for this group. On the opposite, it is clear that the external variables improve the accuracy for households with a more normal consumption, since the total feature group consistently performs better in this case.

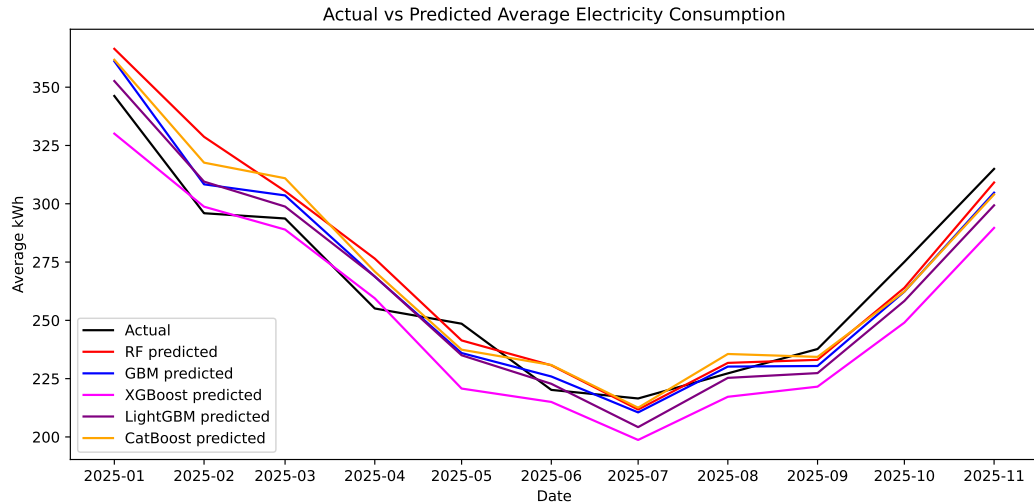


Figure 18: Average actual versus predicted electricity consumption when outlier households (i.e., households with >2000 kWh consumption any month) are removed from the data.

Figure 18 shows the adjusted plot average predicted monthly electricity consumption with the high-consuming households removed. The same pattern of overestimation in the early

year still occurs, although to a lesser extent. The XGBoost model now consistently underestimates the consumption for almost the entire test period. This is confirmed by the error distribution scatter plot in Figure 19, where the regression line for XGBoost has a smaller slope than the perfect prediction line. Remaining models show a more balanced error spread where over- and underestimation is more equally distributed.

TODO: analysera mer om dessa outliers. alla bör ju ha eluppvärmning tex?

The 267 removed households almost exclusively belonged to the house category (89%) and vacation home category (10%), and the majority had electrical heating (70%). Apartment households rarely ever exceed a monthly consumption of 2000 kWh. [something about how we can make the model interpret this better?]

7.2 Explainable AI Forecast Analysis

The results from the XAI analysis of the forecasts provides insights into the ensemble models' decision-making, as well as perspectives on the methodologies used. A notable observation is that the simple feature importance measure did not always rank the variables' importance in the same way as the more detailed SHAP analysis. For example, the feature importance plot for XGBoost ranked the wind-chill-temperature index *WCT* as the most important feature, while the SHAP summary plot showed that several other variables had a more determining impact on the model's predictions. Feature importance in machine learning is characterized by uncertainty and unreliability in some cases. The same features can result in different feature importances depending on the models, FI techniques, and subsets of data used. For non-linear models, feature importances are not fixed, but context dependent, due to variation in feature gradients. Further, the learning process for models used in this study is not the same. For random forest models, the combination och decision tree outputs happens at the end of the training process, while for gradient boosting machines, the outputs are combined at the start [49].

The utilization of SHAP analysis in addition to FI allows for a more elaborate interpretation of the variable contributions. [nånting om att SHAP är bra, även om det också har sina limitations]

One pattern evident in all models is that the calendar information had little impact on the forecasts. Previous works have identified holiday indicators as important for electricity consumption forecasting, since consumption patterns change between weekdays and holidays [39]. Studies on short-term forecasting highlight holidays as anomalies that need to be handled in a special way, for example by introducing holidays features or by modelling holidays in a special way [9, 36, 50]. However, the results in this experiment indicate that monthly forecasts are not affected by holidays in the same way as the short-term forecasts. This could be due to the effect of holidays smoothing out over the entire month, whereas in the case of e.g. day-ahead forecasting, the entire forecast would be either for a holiday or for a weekday.

[comparisons with previous works]

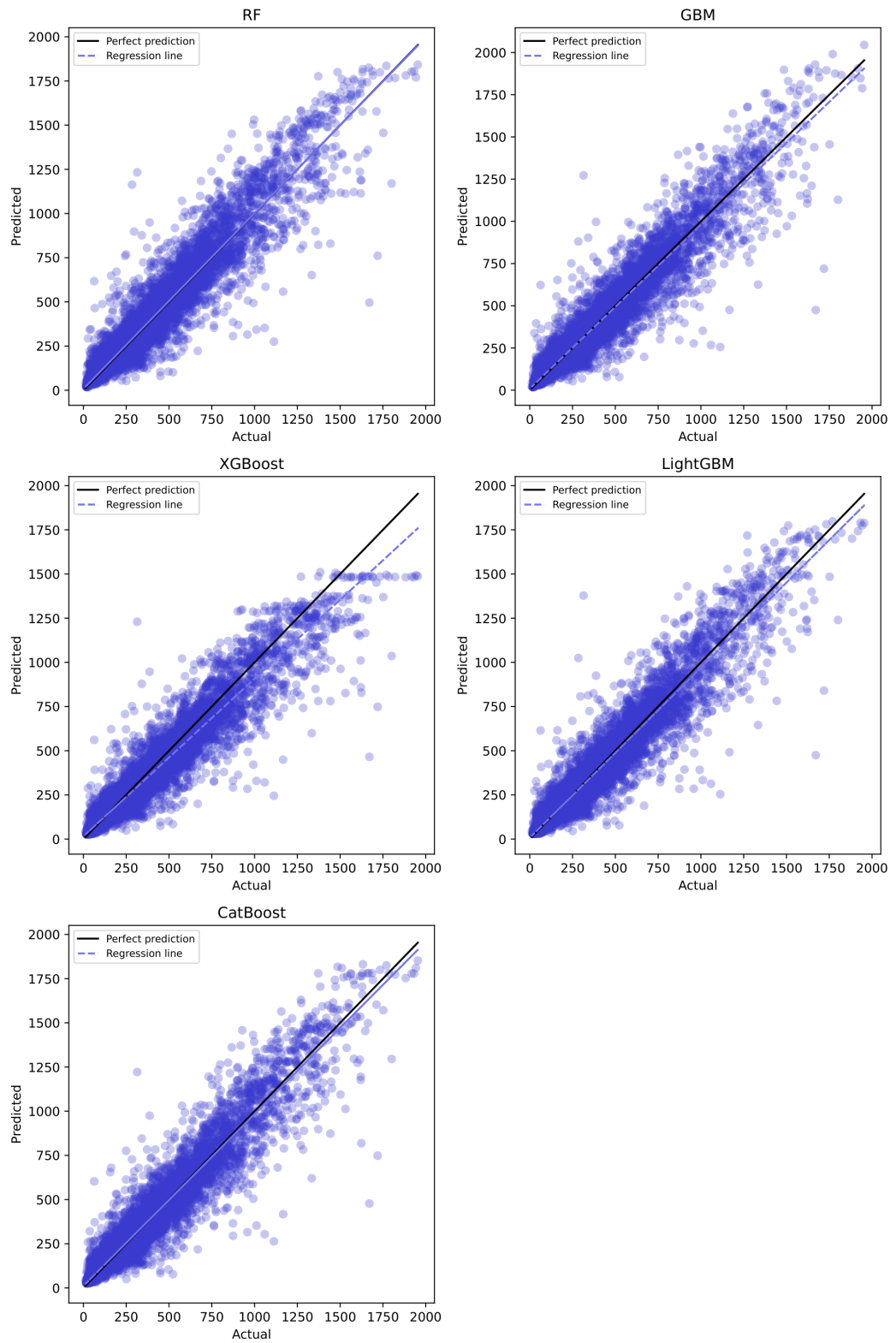


Figure 19: Scatter plots showing prediction error distribution when outlier households (i.e., households with >2000 kWh consumption any month) are removed from the data.

7.3 Limitations

This thesis had some methodological limitations that may have affected the result and how it should be interpreted. Mainly, the large effect on model performance due to outlier values in the electricity consumption [a big limitation is the insufficient handling of outliers, since there were many. This impacted the model performance greatly. But I just did not know what to do lol] [Future research is (probably) needed to better handle the outlier situation of household prediction, since this topic is underexplored in research]

One limitation with this study is the method of constructing certain features, particularly external ones. All weather variables are aggregated to monthly values using mean. Some weather variables may provide more information using another aggregation method, for example, the monthly total amount of solar radiation may be a better variable than the mean solar radiation. Additionally, the process of aggregating hourly values to a single monthly value can discard valuable information. For example, Lazzari et al. [10] found that minimum and maximum temperature were more important than mean temperature for their forecasts, possibly due to minimum and maximum temperatures having a higher variation than the mean. This indicates the existence of potential improvements by creating additional features from the weather variables than just the monthly mean. The same potential exists for the construction of more informational holiday indicators. This study only used the simple number of holidays and total number of days per month, while a better indicator could be categorizing months based on holidays.

Another limitation is the grouping of households performed via clustering. The clusters identified via the K-Means method did not show drastically different properties from each other, and the variation in electricity consumption within each cluster was high, indicated by the high standard deviation present in each cluster. Also, more than 50% of the households belonged to the high consumption profile cluster. The K-Means method may have had difficulties with grouping the data, leading to the cluster feature having a smaller impact than anticipated. More sophisticated techniques may exist to produce a grouping of households that is more informative to the forecasts.

Some limitations were also present in the experimental execution, mostly due to computations being heavily resource-consuming for the large dataset. Firstly, the number of households used for the experiments was limited to a random subset of 15% of the total set of households. Secondly, the grid search for XGBoost was limited in order to speed up the computation, resulting in the optimal parameters in the hyperparameter space possibly not being found. Thirdly, RF used approximate Shapley values in its SHAP analysis, again to speed up computation time. The approximation does not have the same consistency as the full calculation, possibly leading to misleading variable interpretations for this model.

7.4 Real-World Application of the Forecasting System

From a real-world production perspective, the forecasting system proposed in this thesis is limited by assuming full availability of all data, which is not always the case in a real-world scenario. These experiments utilize three years of historical data electricity consumption data, as well as weather data for the upcoming month. Full historical consumption data will not be available for newer customers, creating a need for special solutions when new

customers enter. Weather forecast data from open APIs is available, but rarely for an entire month ahead, making the weather component of the forecasts more uncertain in a production environment.

The choice of decision-tree ensemble models in this thesis alleviates the implementation of the forecast system into a real-world scenario. Decision-tree ensembles are known for their strength in handling uncertain data with outliers or missing values [15, 16]. A robust decision-tree ensemble would therefore be suitable for this application, where some data may be missing or changing. FI and SHAP explanations showed that the models generally prioritized the previous month's consumption over the consumption for the same month last year, indicating that forecasting with these models may be possible even if the amount of historical consumption data is small.

Despite its limitations, the proposed forecasting system has potential to be implemented in a real-life production environment. The robustness and adaptability of the ensemble models is suitable for an environment where the data may be changing, and where some households may not have complete history. The addition of explainable AI analysis to enhance forecast interpretability can also be a valuable perspective for both end-users and the energy company producing the forecasts. For end-users, it helps with understanding what the forecast is based on, which in turn increases its reliability. For the energy company, the explainable AI analysis can guide the developers of the forecasting system to make informed decisions about what data to use for the forecasts. The strength with SHAP analysis is that it can be performed both locally and globally, meaning explanations can be produced for a specified month for an individual household, but also for the general decision-making of the model.

8 Conclusion

[write about how household energy forecasting is special, which the main challenges were during this thesis]

[performance of model]

[write about which variables were the most important]

8.1 Future Work

[address how challenges could be handled, and what directions of research are interesting]

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